Hidden sources of irreproducibility in simulations of hydrophobic assembly

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Abstract

Molecular dynamics simulations are routinely used to explain how hydrophobic molecules, peptides and amphiphiles assemble in water, yet whether the assembled state they report is reproducible is rarely tested. Here we use n-hexane, the simplest hydrophobic solute, as a test case. We ran ten fully independent all-atom simulations of 100 hexanes in water at 310 K for 150 ns each. After 50 ns the runs appeared to disagree: the largest cluster contained 50 to 95 molecules, and the spread between runs was 66 times the statistical error that any single run would report. Continuing the runs showed that the disagreement was one of timing, not of outcome. All ten runs ended as a single droplet of 98–100 molecules, varying by 0.5% between runs over 145–150 ns against 21% at 45–50 ns. But the final droplet was complete anywhere between 33 and 118 ns (median 52 ns), and only half the runs had reached it by 50 ns. The time needed to reach the end state, rather than the end state itself, is what a single trajectory cannot predict, and a run can look converged while it waits for the next droplet fusion. Local structure, conformations and the energy change were reproducible throughout, and early assembly milestones varied far less than the time to the final droplet. Two routine setup steps add further bias. Pressure equilibration contracts the box by 6.8%, raising the concentration from a nominal 200 mM to 215 mM. Surface areas computed without periodic images understate burial by up to 22 percentage points, depending on where a droplet sits. In this model, assembly raises the potential energy by about 940 kJ mol^{-1}, because the hexane–water dispersion lost on burial outweighs the hexane–hexane and water interactions gained. Spontaneous assembly is therefore entropy-driven here, whereas experiments imply a slightly favourable enthalpy for this step at 37 °C. Control simulations trace the difference to the force field, not to the 4-fs time step: liquid hexane is less cohesive in this model than in reality, and moving a hexane from its liquid into water is about 7 kJ mol^{-1} too favourable at 25 °C, whereas runs at 2 fs gave the same assembly and energy change. These results show which quantities a single trajectory can support, and how long it must run before a morphology can be reported, and they motivate a minimal set of checks for simulation studies of assembly.

Introduction

The tendency of non-polar molecules to cluster in water drives protein folding, membrane formation and the assembly of peptide and amphiphile materials[1–4]. Theory and simulation have shown that this hydrophobic driving force depends on length scale. Small solutes are hydrated without disrupting the hydrogen-bond network of water, whereas extended non-polar surfaces cannot be, so the free energy of hydration grows with surface area rather than volume beyond about a nanometre[5–8]. Simulations of hydrophobic polymer collapse and of n-alkane chains in water have made these ideas concrete at the molecular level[9–11].

All-atom molecular dynamics (MD) is now used routinely to simulate the assembly of many molecules at once. In such studies the cluster size, morphology and buried surface area are often taken from one trajectory, or a few, of tens to hundreds of nanoseconds. Assembly from a supersaturated solution, however, is not an equilibrium measurement. It is a coarsening process: nuclei form, grow by monomer addition, and then merge by coalescence and exchange material by ripening[12–15]. The state reached at a given time therefore depends on the history of rare events, such as when two droplets happen to meet, and a trajectory can appear converged while it waits for the next one. The need for independent replicas and careful uncertainty estimates in MD is well recognised[16–21]. In practice, however, error bars are often obtained by block-averaging a single trajectory[18]. This assumes the trajectory has sampled the relevant states, and that assumption fails precisely when the outcome hinges on a few stochastic events.

Two further sources of bias arise in setting up and analysing the simulation rather than in the physics, and both are easy to overlook. First, the concentration specified when a system is built is not the concentration simulated. Solvation tools leave gaps at the box boundary and around solutes, so the box contracts during pressure equilibration. Second, solvent-accessible surface area (SASA) is commonly computed with the Shrake–Rupley algorithm[22] on Cartesian coordinates, as in the widely used MDTraj library[23], without periodic images. Two molecules that touch across a box face are then each counted as exposed. For a single large cluster, the resulting error depends on where the cluster happens to sit.

Here we quantify all three effects in the simplest system that assembles: n-hexane in water. Hexane has no charge, forms no hydrogen bonds and has little conformational complexity, so any irreproducibility cannot be attributed to slow internal rearrangements or specific interactions. We ran ten replicas of 100 hexanes in 150 mM NaCl at 310 K and 1 bar for 150 ns each. Every random input differed between replicas: initial placement, velocities, and integrator and barostat seeds. We first analyse three of the runs in detail as a typical 50-ns study would, and then show what the following 100 ns revealed; the other seven then give the distribution of the time to the end state. We separate the quantities that the replicas reproduce from those they do not, and for each quantity we measure how far the spread between runs exceeds the error a single run would report. We follow every droplet through time to identify the events responsible, and we quantify the two setup artefacts. Finally, we decompose the energy of assembly in this force field into its molecular contributions and test it against the liquid and hydration properties of hexane. This bears on whether the model captures the thermodynamics of the hydrophobic effect at physiological temperature.

Results

A minimal system for hydrophobic assembly

Each replica contained 100 n-hexane molecules, 150 mM NaCl (68 Na^+ and 68 Cl^-) and 24,891–24,929 OPC water molecules[24], for 101,700–101,852 atoms in total (Table 1). Hexane was described with the Amber Lipid21 acyl-chain parameters[25]. In this model every CH_3 and CH_2 group is electrically neutral: carbon carries -0.077 e and each hydrogen $+0.026$ e in CH_3, and -0.023 e and $+0.011$ e in CH_2. Ions used parameters fitted for OPC water[26], and cross interactions followed the Lorentz–Berthelot rules of the Amber force fields. The hexanes were built in the all-trans conformation and placed at random positions and orientations in a 9.40-nm cube, with no two atoms of different molecules closer than 0.5 nm. No contact seen in the trajectories was therefore present at the start. The nominal concentration at this volume is 199.9 mM.

As a check on the hexane parameters, a box of 150 hexanes without water gave a liquid density of 0.6361 ± 0.0003 g mL^{-1} at 298.15 K and 1 bar (20-ns average; Methods), 3% below the experimental 0.655 g mL^{-1} (ref. 27). Its enthalpy of vaporisation, 28.28 ± 0.08 kJ mol^{-1}, is 10% below the experimental 31.56 kJ mol^{-1} (Table 3). Both match the values published with these parameters, 0.636 g mL^{-1} and 28.54 kJ mol^{-1} (ref. 47), so the model is implemented as intended but makes hexane less cohesive than it is. The aqueous solubility of hexane is about 9.5 mg L^{-1} (0.11 mM) at 25 °C[28]. At the simulated concentration of 215 mM (see below), the solution is supersaturated about 1,950-fold, so assembly is thermodynamically inevitable. The question is therefore not whether hexane assembles but how reproducibly simulations capture the process and its outcome.

Simulations used OpenMM 8.4 (ref. 29) on single NVIDIA A100 GPUs, with a 4-fs time step enabled by hydrogen-mass repartitioning[30,31], a Langevin thermostat at 310 K and a Monte Carlo barostat at 1 bar (Methods). Each replica was equilibrated for 0.71 ns and then run for 150 ns of production, at 343–351 ns per day. The mean kinetic temperature was 311.27 ± 0.01 K across the three runs, 1.3 K above the set point. Such an excess is characteristic of discretisation error in Langevin integration at large time steps and is discussed below[32,33]. A box of water and ions alone, run with the same settings for 150 ns (Methods), averaged 311.29 K, so the excess comes from the water and the integrator rather than from hexane. Repeating run 1 at a 2-fs step reduced it to 0.37 K, with or without hydrogen-mass repartitioning (Extended Data Fig. 6c). The mean density of the full system was 0.9899 g mL^{-1}. Temperature was steady throughout, while density, box volume and potential energy drifted only as assembly proceeded, and stopped drifting once each run had formed a single droplet (Extended Data Fig. 1).

Assembly proceeds from a dispersed state by nucleation, growth and droplet fusion

We define a cluster as a set of hexanes connected by heavy-atom contacts within 0.5 nm. In the first production frame, 18, 23 and 25 hexanes were in clusters in runs 1–3, and 18–35 over all ten runs; the largest cluster had two to four members. This is close to random placement of the same molecules at the same volume, which gives 15 ± 5 clustered molecules with a hard-core exclusion and 22 ± 5 without one (see below). Production therefore began from an essentially dispersed solution, and the whole assembly process was recorded (Fig. 1).

Assembly followed the same sequence in all ten runs (Fig. 1b–f). Small clusters formed within the first few nanoseconds. Averaged over the first nanosecond, about a third of the molecules were clustered (30–38 in each run).

The number of clusters with at least two members peaked at 16–19 at 0.8–8.2 ns (0.5-ns running mean; 2.5–3.3 ns in runs 1–3) and then fell as clusters merged and grew (Fig. 1d). We define a stable nucleus as the moment after which the largest cluster never drops below ten molecules (0.5-ns running mean). It appeared at 4.7, 8.9 and 6.3 ns in runs 1–3 and at 2.7–8.9 ns over all ten. The largest cluster then grew by adding monomers and small clusters and passed 30 molecules for good at 16.9, 17.1 and 16.9 ns in runs 1–3. That agreement was partly chance: over all ten runs this milestone ranged from 7.7 to 18.2 ns (Table 2 and Fig. 6c). Ninety per cent of the hexanes were clustered from 37.6, 31.4 and 36.6 ns onwards in runs 1–3 (35.2 ± 3.3 ns) and from 31–65 ns over all ten (40.1 ± 10.0 ns). By the end of the runs the mean size of clusters with two or more members had grown about forty-fold, from about 2.4 to 99 (Fig. 1e).

The runs diverge once droplets begin to coalesce. We call a cluster of at least ten molecules a droplet and follow each droplet through time by its membership. A fusion is a merger of two clusters of at least five molecules each that stays together for at least 0.2 ns (Methods). Runs 1–3 had 2, 3 and 2 fusions in 150 ns. The last fusion in each run joined the final two droplets: 55 + 37 molecules at 43.4–44.6 ns in run 1, 50 + 47 at 55.1 ns in run 2 and 67 + 27 at 93.3 ns in run 3 (Fig. 2c). After it, each run held a single droplet of 94–100 molecules for the rest of the simulation, with the remaining hexanes free in solution (Fig. 1f and Fig. 2b). The largest cluster passed 50 molecules for good at 31.6, 53.8 and 24.6 ns, and 90 molecules at 44.8, 55.2 and 93.5 ns. These times are set by coalescence events rather than by the kinetics of individual molecules, and they vary far more between runs than the nucleation and growth milestones. Runs 4–10 followed the same pattern (Fig. 1b,f and Extended Data Fig. 8), except run 6, in which a single droplet grew to its final size without a droplet fusion.

Runs that disagree at 50 ns converge to the same state

A study that stopped at 50 ns, as ours originally did, would have reported three different end states (Fig. 2a). Averaged over 45–50 ns, the largest cluster contained 94.8, 49.8 and 63.3 molecules, and there were 1.0, 2.1 and 2.1 clusters of two or more members. Each run would have suggested a different morphology: a single droplet, two nearly equal droplets, or one large and one medium droplet. By contrast, the fraction of molecules clustered was nearly the same in all three (94.8, 97.5 and 94.4%). Over all ten runs, the largest cluster contained 50–95 molecules in this window and four runs still held two droplets; even the clustered fraction ranged from 78 to 98%, because in run 6 about a fifth of the molecules were still free (Fig. 1b,c and Table 2).

To measure how misleading a single run would be, we compared two uncertainties for each quantity. The first is the standard deviation between runs. The second is the standard error that one run would report from five 1-ns blocks of a 5-ns window (Methods). If a single run sampled the relevant states, the ratio of the two would be close to 1. Over 45–50 ns, across all ten runs, this ratio is 1.4 for temperature, which serves as a control (Fig. 2d). It is 66 for the largest cluster, 8.4 for the number of clusters, 19 for the buried share of hexane surface (after the correction for periodic images described below) and 20 for the number of free molecules. Runs 1–3 alone give 0.94, 90, 12, 8.4 and 7.4 (diamonds in Fig. 2d), so the conclusion does not depend on which runs are compared. Within each run, the end state looked converged: the block standard error of the largest-cluster size over 45–50 ns averaged 0.25 molecules, against a spread of 17 molecules between runs. A single 50-ns run would thus have reported the size of the largest cluster with an error bar about two orders of magnitude too small.

Continuing the runs resolved the disagreement. Run 2's two droplets merged at 55.1 ns, and run 3's at 93.3 ns (Fig. 2c); the droplets of runs 4–10 are traced in the same way in Extended Data Fig. 8. Over 145–150 ns every run held a single droplet. The largest cluster contained 99.3, 99.0 and 99.5 molecules in runs 1–3 and 98.3–100.0 over all ten (0.5% variation between runs; Fig. 2b), and every frame of every run had exactly one cluster of two or more members, so that quantity no longer varied at all. The standard deviation of the largest cluster between the ten runs fell from 17 molecules at 45–50 ns to 0.49 at 145–150 ns (Fig. 2e). Over 145–150 ns the reproducibility ratio was 3.5 for the largest cluster and 2.9 for the buried share, against 66 and 19 at 45–50 ns, while temperature stayed near 1 (0.85; Fig. 2d). Near convergence the ratio itself becomes unreliable, though. When the runs differ by a single molecule that stays free, or stays bound, for a whole window, a run's block error falls to almost zero. After 120 ns the ratio for the largest cluster ranged from 3.5 to 25 across windows, although the runs differed by only 0.5–1.2 molecules; with runs 1–3 alone it reached more than 800. The absolute spread is then the better guide. Fig. 2e shows when the runs agreed. From 5 ns onwards the spread between runs exceeded the error bar of a single run at least 2.6-fold, and between 30 and 90 ns it was 13–17 molecules, against single-run errors of 0.04–1.3. It fell below 3 molecules only after the last fusion of the slowest run, run 5, at 118 ns, and

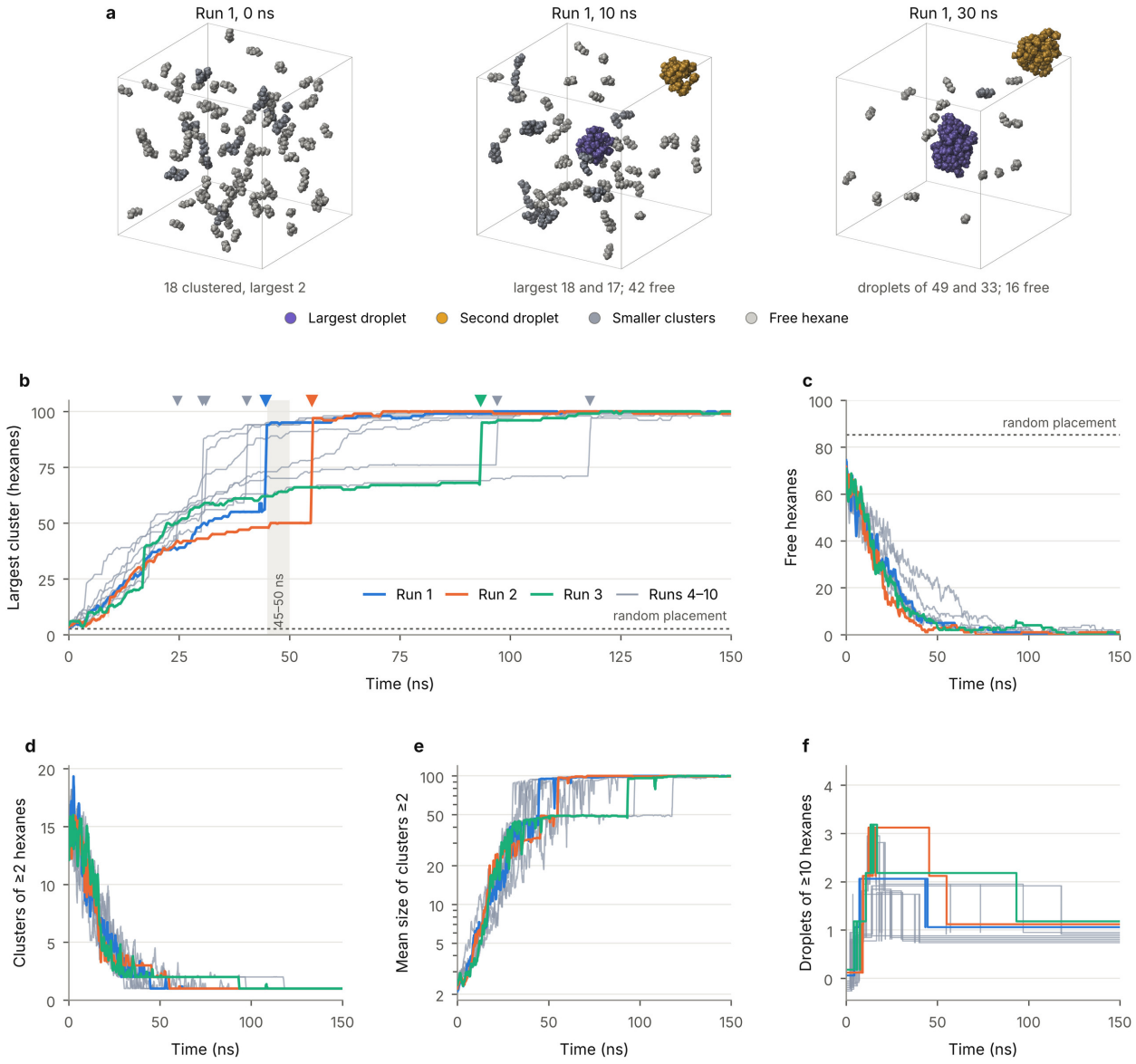


Figure 1 | Assembly of 100 n-hexanes in water in ten independent replicas. a, Run 1 at 0, 10 and 30 ns, drawn as spacefill with water and ions hidden. Each molecule is placed in the box by its centre of mass after the frame is shifted to centre the largest cluster. Each cluster is drawn whole, so part of a cluster can extend past the box outline. Colours mark droplets (clusters of at least ten molecules) by size rank, smaller clusters and free molecules. **b**, Size of the largest cluster against time, as 0.5-ns running means for runs 1–3 (colours) and runs 4–10 (grey); faint lines show every 10-ps frame of runs 1–3. The dashed line is the mean largest cluster for random placement of the same molecules at the production volume (hard-core variant, 30 trials). The shaded band marks 45–50 ns, the end-state window of a 50-ns study, and triangles mark when the last droplet fusion of each run held (grey for runs 4–10; run 6 had none). **c**, Free hexanes, that is, those not in contact with any other hexane, with the random-placement reference (dashed). **d**, Number of clusters with two or more members. **e**, Mean size of clusters with two or more members (logarithmic axis). **f**, Number of droplets of at least ten molecules, at every 10-ps frame; runs 1–3 are offset upwards by 0.06–0.18 and runs 4–10 downwards by 0.05–0.26 so that coinciding lines stay visible. Panels **c**–**e** show 0.5-ns running means. Two hexanes are in contact when any of their carbon atoms are within 0.5 nm. 15,000 frames per run.

below 1 molecule from 135 ns. The end state is therefore reproducible. What varies between runs is the time taken to reach it: in runs 1–3 the last fusion came at 43.4, 55.1 and 93.3 ns, a spread of 64 ± 26 ns (41%), compared with 10% for the time to cluster 90% of the molecules. Ten runs give a fuller picture (see below).

Why does the last fusion take so long, and why is its timing so variable? The final two droplets of runs 1–3 coexisted for 36, 46 and 83 ns before they merged, counting from when both first reached ten molecules. For most of that time they were far apart: their closest carbon–carbon distance had a median of 4.0, 2.9 and 3.3 nm, and they were within 1 nm of each other in only 0.3, 1.9 and 0.3% of frames (Extended Data Fig. 5). In run 1 the droplets merged on their first close approach. In run 3 they came within 1 nm of each other three times between 20 and 23 ns without merging, and not again until the approach that merged them at 93.3 ns. In run 2 they approached 14 times from 36.7 ns onwards, as close as 0.61 nm, and only the last approach merged them; contact is defined at 0.5 nm. Molecules also passed between the two droplets through the solution only rarely: 10, 11 and 12 times, about one every 4–7 ns. In run 3, 10 of the 12 went from the smaller droplet to the larger one, the direction expected for Ostwald ripening[13,14], but at this rate ripening could not have removed the smaller droplet within the simulated time. The end state was reached by coalescence, after a waiting time set by how slowly whole droplets diffuse (see below). A Smoluchowski estimate for two droplets of the observed sizes and diffusion coefficients that merge on contact in a box of this volume gives a mean waiting time of 78–100 ns (Methods), the same order as the observed 55 ± 25 ns.

Local structure and the apparent potential of mean force are reproducible

The centre-of-mass RDF of hexane, averaged over each 150-ns run, has a sharp first peak at 0.5724 ± 0.0002 nm with a height of 31.0 ± 0.6, and a first minimum at 0.878 nm (Fig. 3a). The first shell holds 8.33 ± 0.20 neighbours, and each hexane is in heavy-atom contact with 6.43 ± 0.14 others on average. The three curves are almost indistinguishable, and they were already so over the first 50 ns, when the runs differed in morphology. Local packing is therefore the same whether a molecule sits in a droplet of 99 or of 31. The height of the peak, however, depends on how long the run is: over 0–50 ns it was 21.6 ± 0.5. A whole-run average weights the dispersed and the condensed states by the time spent in each, so it can be reproducible between runs of the same length without being converged.

Converting g(r) to $w(r) = -kT \ln g(r)$ gives a well of -8.85 ± 0.05 kJ mol^{-1} (-3.43 kT) at 0.572 nm. A barrier of 1.62 ± 0.04 kJ mol^{-1} separates it from the second shell (Fig. 3b). This w(r) should not be read as the pair potential of mean force for two hexanes in dilute solution. Here g(r) is normalised by the mean number density of hexane in the box, but for most of each run nearly all the molecules sit in droplets whose internal density is far higher; for liquid hexane in this model the ratio is about 34. The depth of the well therefore reflects mainly the density of the droplets relative to the box and the time spent condensed, not the free energy of forming one contact. Like the RDF, it deepened from -7.92 kJ mol^{-1} over 0–50 ns as the runs were extended. What matters here is that this structural fingerprint is reproducible between runs to within 0.5% in depth and 0.03% in position.

Molecular conformations are just as reproducible. Of the three C–C–C–C dihedrals in each hexane, 42.9% are gauche ($|\phi| < 120°$) in clustered molecules and 45.8–46.2% in free ones over 150 ns, across the three runs. A free hexane in water is thus slightly more compact than one inside a droplet, and all three runs resolve this small difference identically.

Assembly is energetically uphill in this model

From the first 5 ns to the last 5 ns, the potential energy of the system rose by 983, 940 and 932 kJ mol^{-1} (952 ± 27 kJ mol^{-1}). The three values agree within their uncertainties of 53–79 kJ mol^{-1}, obtained by combining the standard errors of the two 5-ns means from 1-ns blocks. The rise was not steady. Over the first 50 ns it tracked the buried hexane surface (Pearson r = 0.93–0.96 over 1-ns blocks, 5.6 kJ mol^{-1} per nm^2) and reached 1,057 ± 32 kJ mol^{-1} at 45–50 ns. After that the buried surface grew by another 12–29 nm^2, mostly through the final fusions, but the potential energy did not rise further (Extended Data Fig. 1d). Across the final fusion itself it fell by 38, 88 and 129 kJ mol^{-1} in runs 1–3 (10-ns windows on either side; standard errors 40–54 kJ mol^{-1}), while burying 29, 18 and 11 nm^2 more surface. Over all 150 ns the correlation with burial is therefore weaker (r = 0.82–0.87), and the pooled slope is 4.1 kJ mol^{-1} nm^{-2} (r = 0.85; Fig. 3c). The box volume grew by 2.0–2.1 nm^3. At 1 bar the corresponding pressure–volume work is only about 0.1 kJ mol^{-1}, and the thermostat holds the kinetic energy fixed. The enthalpy of assembly is therefore essentially the change in potential energy: about +950 kJ mol^{-1} for the system of 100 hexanes, or +4.5 kJ mol^{-1} per nm^2 of buried surface (about

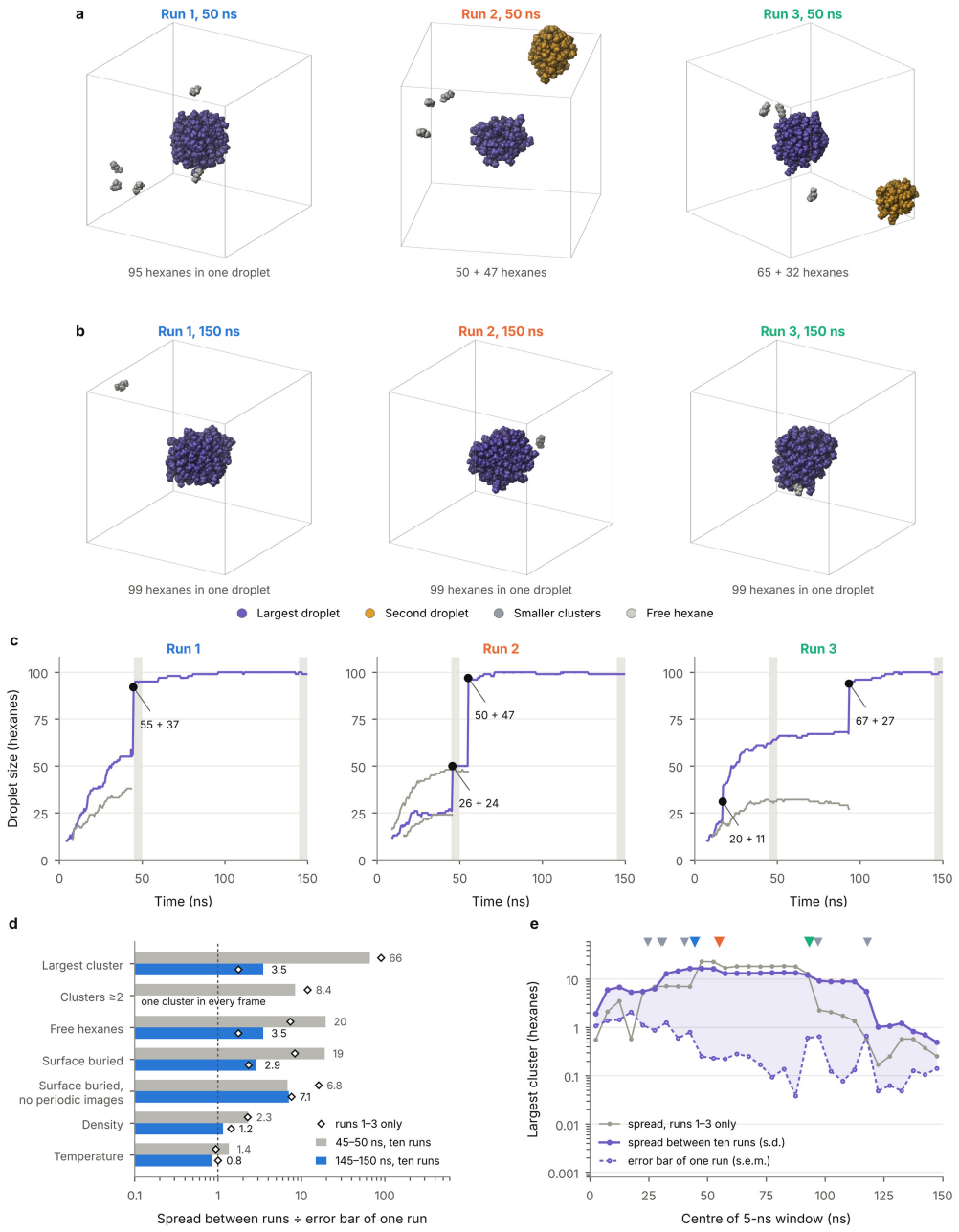


Figure 2 | Runs that differ at 50 ns reach the same state at different times. a, Runs 1–3 at 50 ns and **b**, at 150 ns, drawn as in Fig. 1a; views at 50 ns were chosen so that separate droplets do not overlap in projection. **c**, Size of every droplet (cluster of at least ten molecules) in runs 1–3 against time, each droplet followed by its membership (0.5-ns running mean); runs 4–10 are shown in Extended Data Fig. 8. Purple lines are the droplet present at 150 ns and grey lines droplets that merged into another. Dots mark fusions between two droplets, labelled with their sizes just before contact. Shaded bands mark 45–50 and 145–150 ns. **d**, Standard deviation between the ten runs divided by the standard error of the mean that a single run reports from five 1-ns blocks, for 45–50 ns (grey) and 145–150 ns (blue), averaged over runs; diamonds give the same ratio for runs 1–3 alone. A value near 1 (dashed line) means that one run estimates its own uncertainty correctly. Over 145–150 ns every frame of every run has exactly one cluster, so no ratio is defined for the number of clusters; surface quantities are computed with periodic images unless labelled otherwise. **e**, Standard deviation of the largest cluster between the ten runs (purple) and between runs 1–3 alone (grey), and the mean standard error that a single run reports (dashed), in every 5-ns window, plotted at the window centre (logarithmic axis). Shading marks where the spread between the ten runs exceeds the error bar of one run. Triangles mark when the last droplet fusion of each run held (grey for runs 4–10).

+14 kJ mol^{-1} for each molecule's worth, taking 3.03 nm^2 as the SASA of an isolated hexane). Assembly is spontaneous ($\Delta G < 0$) even though $\Delta H > 0$, so the entropy change must be positive and exceed $\Delta H/T$. In this model, assembly at 310 K is thus entropy-driven, the classic signature of the hydrophobic effect near room temperature[1,2].

To see where this cost comes from, we split the potential energy exactly into contributions from pairs of molecule types, re-evaluating saved full-system frames with the force field used in the simulation (Methods). The recomputed energies matched the logged ones to within 1.8 kJ mol^{-1} out of 1.3×10^6, and the pair terms summed to the total to within 0.01 kJ mol^{-1}. Between 0–5 and 145–150 ns, assembly gives up 3,880 ± 9 kJ mol^{-1} of hexane–water interaction, almost all of it Lennard-Jones dispersion. It recovers 1,741 ± 3 kJ mol^{-1} from new hexane–hexane contacts (Fig. 3d and Extended Data Fig. 3). Hexane's own bonded and intramolecular energy changes little (−84 kJ mol^{-1}), and so do its contacts with ions (+31 kJ mol^{-1}). Water and ions recover 1,134 ± 10 kJ mol^{-1}, obtained as the logged change in potential energy minus the hexane terms. The direct estimate from the saved frames, −1,313 kJ mol^{-1}, agrees within its larger uncertainty. Both hexane terms scale with the buried surface: by +18.7 kJ mol^{-1} nm^{-2} for hexane–water and −8.3 kJ mol^{-1} nm^{-2} for hexane–hexane over all analysed frames (Fig. 3e). Within each run, the changes divided by the change in buried surface are 18.3–18.4 and −8.3 to −8.2 kJ mol^{-1} nm^{-2}. In this model, then, a buried hexane loses more dispersion attraction to water than it gains from its neighbours, and the water released from its hydration shell makes up only part of the difference. The levelling of the potential energy after 50 ns sits in the water-and-ion term. Across the final fusion of each run it fell by 450, 301 and 280 kJ mol^{-1}, more than during growth for the same burial, while the hexane terms rose by 412, 213 and 151 kJ mol^{-1}, in line with the extra burial; afterwards it stayed level to within about 77 kJ mol^{-1} until 150 ns. It changed from −793 ± 147 kJ mol^{-1} at 45–50 ns to −1,134 ± 10 kJ mol^{-1} at 145–150 ns. Merging two droplets therefore releases more energy from the surrounding water, per unit of surface buried, than adding molecules one at a time; we do not resolve the cause here.

Experiment suggests the balance should already have shifted by 37 °C. Near room temperature, the enthalpy of transferring a liquid hydrocarbon from water to its own liquid is close to zero. Because hydrophobic hydration has a large positive heat capacity, the transfer becomes increasingly exothermic as the temperature rises[34,35]. Baldwin's analysis puts the temperature of zero transfer enthalpy for liquid hydrocarbons near 22 °C[34], so a modestly favourable enthalpy of aggregation is expected at 37 °C. The positive enthalpy found here has the opposite sign. The decomposition narrows down where the discrepancy could arise. The hexane terms follow burial directly and are measured precisely, so they cannot be a drift. They depend on the hexane–water cross interaction, which here comes from the Lorentz–Berthelot combining rules (carbon–oxygen ε = 0.638 kJ mol^{-1}, σ = 0.328 nm) rather than from a fit to alkane hydration data. The water-and-ion term is obtained by difference, so it could in principle contain a slow drift unrelated to assembly. A hexane-free control rules this out. The same box of water and ions, run for 150 ns with the same protocol, drifted by −0.01 ± 0.22 kJ mol^{-1} per ns, or −1 ± 32 kJ mol^{-1} over the 145 ns separating the windows used here, and changed by −8 ± 73 kJ mol^{-1} between 0–5 and 145–150 ns (Extended Data Fig. 1d). Both are consistent with zero and under 3% of the water-and-ion term. Nor is the time step responsible: run 1 repeated at 2 fs gave a rise of 968 ± 92 kJ mol^{-1} with hydrogen-mass repartitioning and 993 ± 90 kJ mol^{-1} without it, against 952 ± 27 kJ mol^{-1} for the three runs at 4 fs (Extended Data Fig. 6b).

The cross interaction, by contrast, accounts for the sign. In this model the hydration enthalpy of a single hexane is −35.0 ± 1.3 kJ mol^{-1} at 25 °C, so moving a hexane from its liquid into water releases 6.7 ± 1.3 kJ mol^{-1} (Table 3), where experiment gives close to zero[34]. About half of this error is the 3.3 kJ mol^{-1} shortfall in the enthalpy of vaporisation, which the authors of these parameters also report for short alkanes[47]. The other half is a hydration enthalpy about 3.5 kJ mol^{-1} more negative than experiment implies. At 310 K the model's transfer enthalpy is still negative (−4.0 ± 1.4 kJ mol^{-1}), whereas experiment implies a positive value. An error of about 7 kJ mol^{-1} per molecule is comparable to the rise of 9.5 kJ mol^{-1} per hexane that accompanies assembly here, so the sign of the assembly enthalpy is a property of this force field rather than of hexane.

Molecules stay mobile, but whole droplets move slowly

Fitting the mean-squared displacement (MSD) of hexane centres of mass over lag times of 1–10 ns gives diffusion coefficients of 0.365, 0.352 and 0.378 nm^2 ns^{-1} over the first 50 ns (0.365 ± 0.013 nm^2 ns^{-1}; Fig. 4a,b). Over 50–100 ns the values are 0.091, 0.074 and 0.122, and over 100–150 ns, when every run holds a single droplet, 0.052, 0.081 and 0.085 nm^2 ns^{-1}. The apparent diffusion coefficient, MSD/6t, falls with lag time in every window (Fig. 4a). Over short

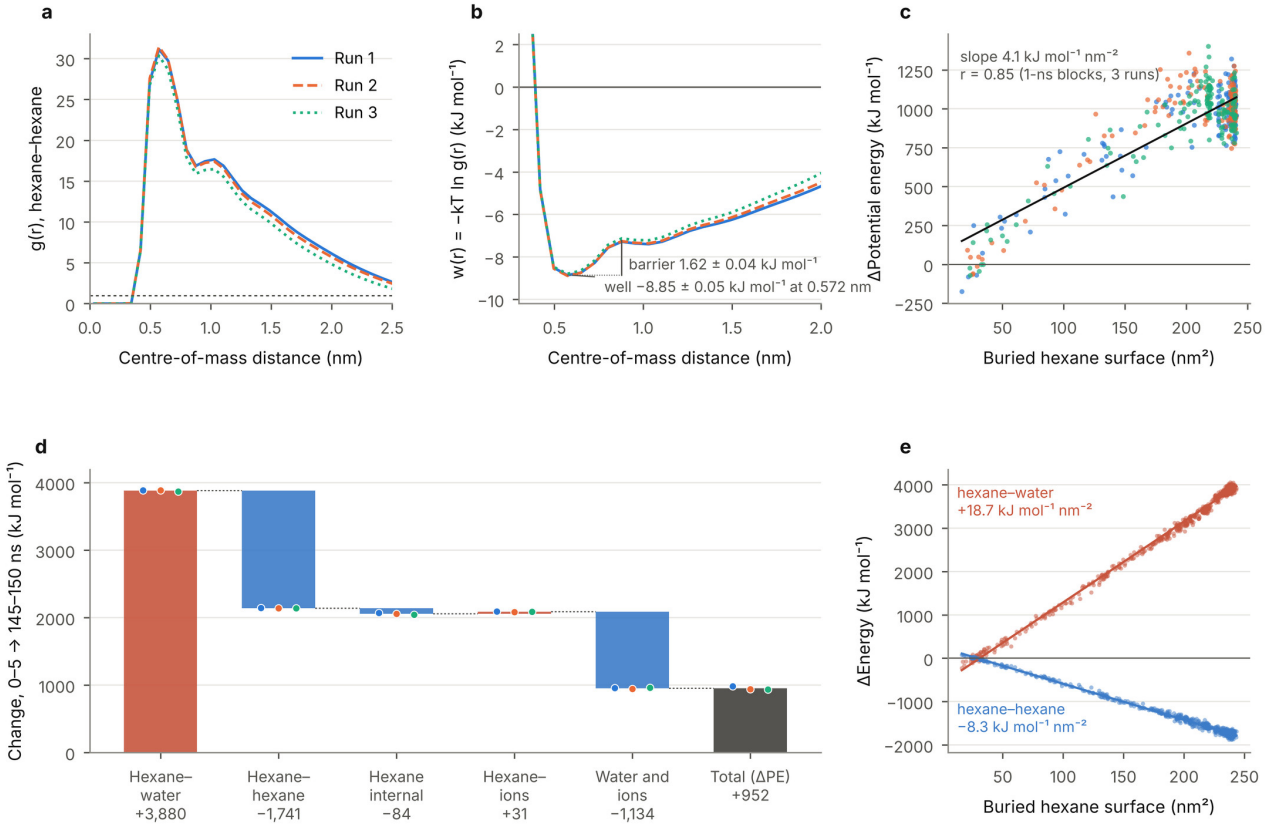


Figure 3 | Structure and energetics of assembly. **a**, Centre-of-mass radial distribution function of hexane, averaged over each 150-ns run (15,000 frames, minimum-image convention, 0.077-nm bins). **b**, Apparent potential of mean force, w(r) = −kT ln g(r) at 310 K, annotated with the depth and position of the well and the barrier to the second shell (mean ± s.d. over runs). Because g(r) is normalised by the mean hexane density in the box, w(r) reflects mainly the density of the condensed droplets and is not a pair potential for dilute solution. **c**, Change in potential energy against buried hexane surface (with periodic images) for 1-ns blocks of all three runs (450 points). Potential energy is relative to the 0–5 ns mean of each run, and the line is a least-squares fit. **d**, Exact decomposition of the change in potential energy between 0–5 ns and 145–150 ns, drawn as a waterfall from zero to the total. Bars show means over runs and dots show the individual runs; red steps raise the energy and blue steps lower it. Hexane terms come from the ten saved full-system frames in each window, and the water-and-ion term is the logged change in potential energy minus the hexane terms. **e**, Hexane–water and hexane–hexane interaction energies, relative to 0–5 ns, against buried hexane surface for every analysed full-system frame of all three runs (615 points), with least-squares lines.

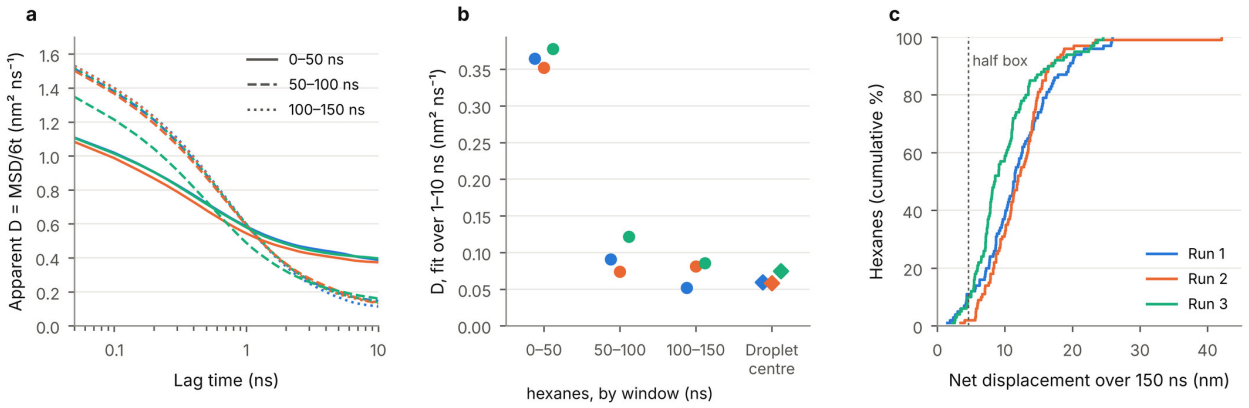


Figure 4 | Hexanes remain mobile, but droplets move slowly. a, Apparent diffusion coefficient, MSD/6t, of hexane centres of mass against lag time (logarithmic axis) over 0–50 ns (solid), 50–100 ns (dashed) and 100–150 ns (dotted); colours identify runs 1–3. MSDs were averaged over molecules and time origins within each window (frames every 50 ps, coordinates unwrapped across periodic boundaries). **b**, Diffusion coefficients from linear fits of the MSD over 1–10 ns: hexanes in each 50-ns window (circles), and the centre of the single droplet after the last fusion of each run (diamonds). **c**, Cumulative distribution of the net centre-of-mass displacement of each hexane over 150 ns. The dashed line marks half the box edge.

times a hexane moves freely in water or inside a liquid-like droplet, but over longer times most molecules are carried by droplets that themselves diffuse slowly. The long-time coefficient is therefore an average over free and droplet-bound motion, not the diffusion coefficient of a hexane monomer, and it falls as assembly proceeds. It is also not corrected for the finite size of the box[36]. The time-resolved values show the same pattern as the morphology: run 3, which held two droplets until 93 ns, kept the highest value over 50–100 ns.

Over 150 ns, 89, 98 and 90 of the 100 hexanes travelled a net distance greater than half the box edge (4.6 nm), and the median net displacement was 8.5–11.9 nm (Fig. 4c). Over 150 ns every hexane came within 1.5 nm (centre of mass) of every one of the other 99 molecules at least once, against 76–96 partners on average over the first 50 ns. The end-to-end vector of each molecule decorrelated to 1/e within the first sampled lag of 50 ps. The assembled state is therefore liquid-like at the molecular level. The irreproducibility at 50 ns did not come from individual molecules being kinetically trapped; it came from slow coarsening at the level of whole droplets. Droplets of 21–40 molecules followed before their last fusion diffused at 0.12–0.14 $nm^2\ ns^{-1}$, and the single droplet of about 99 molecules at 0.060, 0.058 and 0.075 $nm^2\ ns^{-1}$ (Fig. 4b and Methods), five to six times more slowly than the average hexane over the first 50 ns. Treated as a sphere of liquid hexane with stick boundary conditions, a droplet of 99 molecules (radius 1.75 nm) is predicted by the Stokes–Einstein relation, with the viscosity of water at 37 °C (0.69 mPa s[27]), to diffuse at 0.19 $nm^2\ ns^{-1}$ in an unbounded solvent. The hydrodynamic finite-size correction for a 9.2-nm box removes 0.10 $nm^2\ ns^{-1}$ of that[36], leaving 0.09 $nm^2\ ns^{-1}$, close to what we measure.

Equilibration silently changes the concentration

The box was built as a 9.40-nm cube (830.6 nm^3), corresponding to 199.9 mM hexane. During pressure equilibration it contracted to 773.8 ± 0.7 nm^3 (edge 9.181 nm), losing 6.8% of its volume. Apart from a growth of about 2 nm^3 that accompanied assembly (Extended Data Fig. 1), it stayed at that size throughout the 150 ns of production. The simulated concentration was therefore 214.6 mM (18.5 mg mL^{-1}), 7.3% above the nominal value. The contraction most likely arises because the solvation step leaves gaps around each solute and at the box faces, which the barostat then closes. Its size depends on the number and size of the solutes and on the solvation tool, so a concentration quoted from the build can be off by several per cent. Here the shift matters little, because the solution is supersaturated about 1,950-fold either way. It would matter more in studies that compare concentrations or work near a critical aggregation concentration. Reported concentrations should be computed from the mean production volume. Repeating the equilibration of each box with every stage logged shows when this happens (Extended Data Fig. 7). The box kept its built volume through minimisation and heating at constant volume, then contracted by 7.0–7.1% in the first constant-pressure stage, while the hexanes were

still restrained at 1,000 kJ mol^{-1} nm^{-2}. Half of the contraction took 8–9 ps after the barostat was switched on and 90% took 26–32 ps; the volume then fluctuated about its final value for the remaining 0.5 ns. The density rose from 0.922 to 0.992 g mL^{-1}, so the built box was about 7% short of water.

A second check concerns the starting state. In common protocols, including the one used here, no configurations are saved during equilibration, so assembly that begins before production would go unnoticed. To test for this, we placed the same 100 molecules at random in a box of the production volume (773.1 nm^3) 30 times. Each placement was done both with and without a hard-core exclusion of 0.3 nm between heavy atoms of different molecules (Methods). With the hard core, random placement gives 14.8 ± 4.6 clustered molecules, a largest cluster of 2.6 molecules and 0.16 contacts per molecule. Without it, the corresponding values are 21.7 ± 5.1, 3.2 and 0.24. The first production frames of runs 1–3 (18, 23 and 25 clustered molecules; largest cluster 2–3) lie within these distributions (Fig. 5a). Over all ten runs the first frames held 18–35 clustered molecules. Runs 5, 8 and 10 (31, 35 and 35) lie 1.8–2.6 standard deviations above the distribution without a hard core, so some contacts may have formed during the final unrestrained stage of equilibration, but no run started with a cluster of more than four molecules. The 0.71-ns equilibration therefore did not pre-assemble the solution, and the recorded trajectories capture the process from its beginning. We recommend this comparison as a routine check. For molecules that associate more strongly, or systems that need longer equilibration, substantial clustering during equilibration would otherwise go unreported.

Surface areas computed without periodic images depend on where the droplet sits

A widely used way of computing SASA is the Shrake–Rupley routine of MDTraj[22,23], which works on Cartesian coordinates and ignores periodic images. Molecules are kept whole, so two hexanes that touch across a box face are each counted as exposed at that contact. For a solution of monomers this barely matters. For a droplet of tens of molecules in a 9.2-nm box, however, part of the droplet often lies across a face.

To measure the effect, we built one compact droplet of 95 hexanes with centres within 1.8 nm of each other. We placed it in a 9.18-nm box at the centre, on a face, on an edge and on a corner (Methods). Without periodic images, the buried share of its surface was 59.8%, 51.8%, 45.2% and 37.7% in these four positions. With the atoms of the 26 neighbouring images that lie within 0.8 nm of a face included, it was 59.8% in every position (Fig. 5b). The same droplet can therefore appear up to 22 percentage points less buried, depending only on where it sits. That range is larger than the spread of the uncorrected buried share between our three runs at 45–50 ns (56.5%, 65.8% and 73.5%; Fig. 5c).

We therefore recomputed every surface area in all ten runs with periodic images (Methods). The correction is not constant. The surface missed without images varies between 0 and 87 nm^2 as droplets move relative to the box faces, and averages 7–24 nm^2 over each run (Fig. 5d and Extended Data Fig. 4, for runs 1–3). At 45–50 ns, the corrected buried share of runs 1–3 is 74.6, 71.1 and 69.2%, instead of 73.5, 65.8 and 56.5% (Fig. 5c). For these three runs the spread shrinks from 17.0 to 5.4 percentage points, the CV from 13% to 3.8% and the reproducibility ratio from 16 to 8.4, so about two-thirds of their apparent variation in burial at 50 ns was an artefact of where the droplets happened to sit. Over all ten runs, which genuinely differed at that time, the artefact adds less: the standard deviation of the buried share is 5.1 percentage points with periodic images and 5.6 without (Table 2). The final state makes the point more sharply. Over 145–150 ns every run held a single droplet of 98–100 molecules, and the corrected buried share was 77.7–79.0% across the ten runs (CV 0.5%). Without periodic images it was 65.9–78.6% (CV 5.9%), because the same droplet sat at different places relative to the box faces; in runs 1–3 the surface missed averaged 6.5, 3.4 and 32.5 nm^2 in this window. The potential energy, which is computed with full periodicity, pointed the same way: it rose by similar amounts in all runs (940 ± 65 kJ mol^{-1}). The artefact is worse if coordinates are wrapped into the box atom by atom instead of molecule by molecule, because molecules are then split across faces. In our random-placement test, atom-wise wrapping made the computed buried surface of a dispersed solution negative.

Ten runs give the distribution of the time to the final droplet

To estimate how long the end state takes to reach, we ran seven further replicas (runs 4–10) from independent builds with the same settings (Methods). All ten runs ended as a single droplet. Over 145–150 ns the largest cluster contained 99.4 ± 0.5 molecules (range 98.3–100.0), a variation of 0.5% between runs, and every frame of every run held exactly one cluster (Fig. 6a,b and Table 2). The route there varied. Nine runs completed the assembly by merging their last two

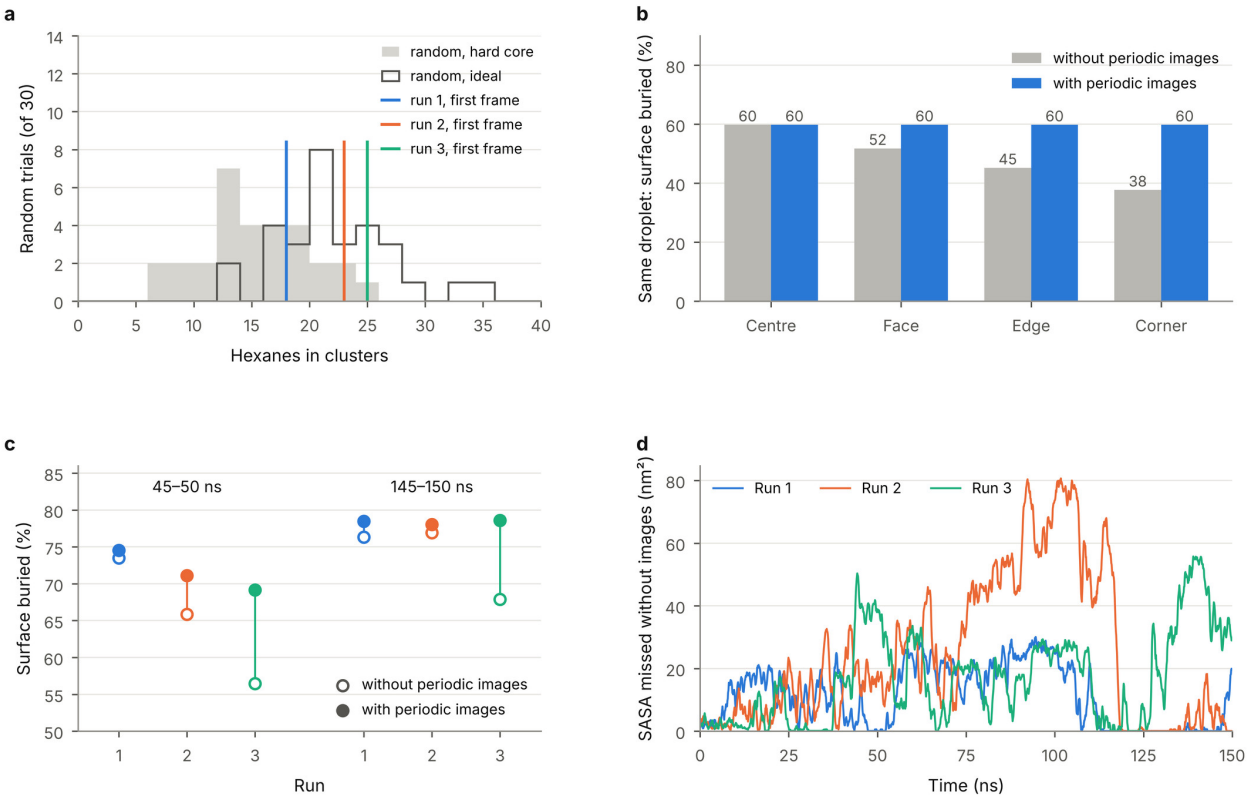


Figure 5 | Two setup checks: the starting state and periodic surface area. a, Hexanes in clusters in the first production frame of each run (vertical lines), compared with 30 random placements of the same molecules at the production volume. Placements used a 0.3-nm hard-core exclusion between heavy atoms of different molecules (filled histogram) or no exclusion (open histogram). **b**, Buried share of the surface of one compact droplet of 95 hexanes (centres within 1.8 nm), placed at the centre, on a face, on an edge or on a corner of a 9.18-nm periodic box. It was computed without (grey) and with (blue) periodic images. **c**, Buried share of hexane surface in each run over 45–50 ns and 145–150 ns, computed without (open circles) and with (filled circles) periodic images. **d**, Surface area missed when periodic images are ignored: non-periodic minus periodic SASA, every 0.1 ns (0.5-ns running mean).

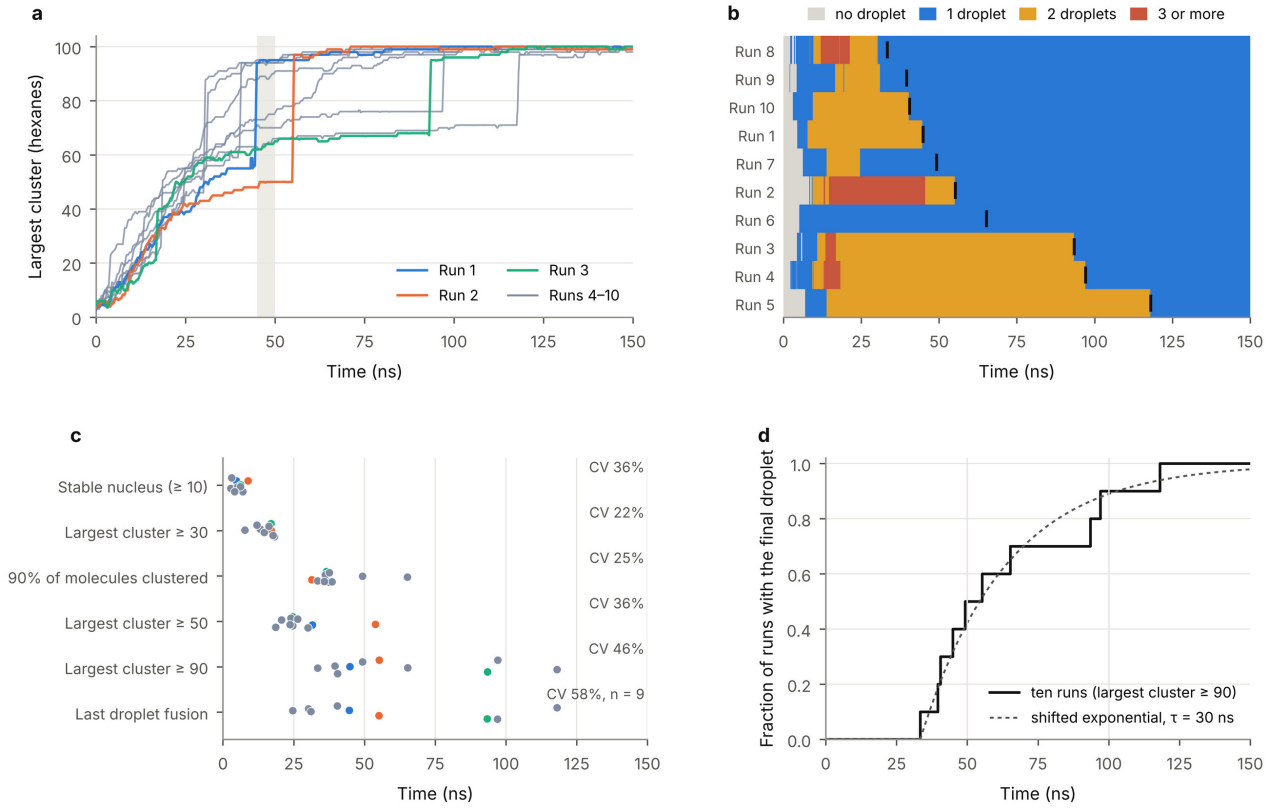


Figure 6 | Ten independent runs reach the same droplet at widely different times. a, Largest cluster in runs 1–3 (colours as in Fig. 1) and runs 4–10 (grey), as 0.5-ns running means; the band marks 45–50 ns. **b**, Number of droplets (clusters of at least ten molecules) in each run against time, runs ordered by when the final droplet was complete (black ticks: the largest cluster reaches 90 molecules for good). **c**, Assembly milestones in each run, with their coefficient of variation across runs; colours as in a. Run 6 had no droplet fusion, so the last row has nine points. **d**, Fraction of runs in which the largest cluster has reached 90 molecules for good (solid), with a shifted exponential fitted by maximum likelihood (dashed; onset 33 ns, mean wait 30 ns beyond it).

droplets, at 25–118 ns, after the two had coexisted for 7–83 ns. In run 6 a single droplet formed at 5 ns and grew to 90 molecules by 65 ns by absorbing monomers and small clusters, without ever meeting a second droplet.

Measured by when the largest cluster first held 90 molecules for good, the final droplet was complete at 33–118 ns, with a median of 52 ns (Fig. 6d). Five of the ten runs had reached it by 50 ns, seven by 75 ns, nine by 100 ns and all ten by 118 ns. These times are consistent with a shifted exponential, with an onset at 33 ns and a mean wait of 30 ns beyond it (Kolmogorov–Smirnov P $\approx$ 0.9), as expected for a waiting time set by a rare event. At 45–50 ns, the window a 50-ns study would report, the largest cluster contained 79 ± 17 molecules (50–95), and four runs still held two droplets. The spread between runs was 66 times the standard error that a single run would report, close to the value of 90 found with three runs.

The early milestones varied far less in absolute terms. A stable nucleus formed at 2.7–8.9 ns and the largest cluster passed 30 molecules at 7.7–18.2 ns, whereas the time to the final droplet had a standard deviation of 29 ns (Fig. 6c). The time to cluster 90% of the molecules was 31–38 ns in eight runs but 49 and 65 ns in the other two, in which the last free molecules and small clusters joined only as the final droplet formed; with three runs it had appeared tighter than it is. The rise in potential energy reproduced: 940 ± 65 kJ mol^{-1} over all ten runs, a spread equal to the error that a single run reports (ratio 0.94).

Discussion

Even for the simplest hydrophobic solute, independent simulations agree on how hexane assembles and on the local structure it forms. At 50 ns they did not agree on what it had assembled into, and a further 100 ns showed that they did not need to: all ten runs reached the same single droplet, at times from 33 to 118 ns. The division between what

reproduces early and what does not follows a physical line. Some properties are set by local interactions and averaged over many molecules: packing, coordination, conformation, short-time mobility, the energetic cost of assembly and, once computed correctly, the buried surface. These converge quickly because each frame already averages over many molecular environments. Other properties count droplets: the largest cluster, the number of aggregates and the number of free molecules. These depend on a handful of coalescence events, so a trajectory samples them only once, and they agree between runs only after every run has completed its last event. Ten runs show that the line is less sharp than three suggested. The fraction of molecules clustered and the time to cluster 90% of them were reproducible in most runs, but not in two in which the last free molecules joined only as the final droplet formed.

After nucleation, the system lowers its free energy by reducing the total area of its hexane–water interface. It does so by coalescence when droplets meet[12] and by ripening, in which molecules leave small droplets for large ones[13,14]. Ripening was too slow to matter here: about one molecule every 4–7 ns passed between the last two droplets of each of runs 1–3. Coalescence completed the assembly in nine of the ten runs, but only after the last two droplets had found each other, which took 7–83 ns (median 28 ns); in the tenth, a single droplet grew to its final size without meeting another. Whole droplets diffuse slowly: 0.12–0.14 $nm^2\ ns^{-1}$ for 21–40 molecules and 0.06–0.07 $nm^2\ ns^{-1}$ for the final droplet of 99, in line with the Stokes–Einstein relation once the finite-size correction for the periodic box is included[36]. For two droplets of the sizes in runs 1–3 merging on contact in this box, the Smoluchowski rate[12] gives a mean waiting time of 78–100 ns, the same order as observed in those runs; in several of the other runs the second droplet was smaller and the wait shorter. Droplets that do approach can also fail to merge, as in run 2, where the last two droplets came within 1 nm of each other 13 times before they merged. The last fusion is therefore a single stochastic event whose waiting time is comparable to the length of a typical simulation, and over ten runs the times to the final droplet are consistent with a shifted exponential (Fig. 6d). At this supersaturation the equilibrium state in a periodic box of this size is a single droplet, and all ten runs reached it. The state at 50 ns was a snapshot of a non-stationary process in which four of the ten runs still held two droplets. Because the box itself slows droplet diffusion, how far coarsening gets within a given time also depends on box size.

This has direct consequences for error estimates. Block averaging assumes that blocks are nearly independent[18]. Before the last fusion, the configurations that matter for morphology are correlated over the waiting time for that fusion, which can exceed the run. Single-run standard errors for morphological quantities at 50 ns were therefore 8 to 66 times too small over ten runs (7 to 90 times with runs 1–3 alone). The ratio of the between-run standard deviation to the single-run standard error is a simple diagnostic for this. It is about 1 for well-sampled quantities such as temperature and far above 1 when a run has not sampled the relevant states. Once every run had completed its last fusion, the spread between runs fell to about one molecule or less. Near such a floor the ratio becomes noisy, because a run's block error can vanish when nothing changes within a window, so it should be read together with the absolute spread. With three replicas the standard deviation is itself uncertain; its 95% confidence interval spans roughly 0.5 to 6 times the estimate. Even the smallest of the 50-ns end-state ratios, 7.4, has a 95% interval of about 4 to 47, well above 1, whereas the temperature ratio of 0.94 is consistent with 1. With ten replicas the interval narrows to 0.7–1.8 times the estimate, and at 45–50 ns the ratios remain far above 1, 66 for the largest cluster and 20 for the number of free molecules, against 1.4 for temperature (Table 2). A single trajectory cannot reveal by itself that it has stopped early. Run 3 held two droplets whose sizes changed by fewer than ten molecules between 30 and 93 ns, and any window in that stretch would have looked converged; run 5 held two droplets from 55 ns until they merged at 118 ns. Two practical remedies follow. One is to run each replica past its last fusion, which can be checked directly by following the number of droplets until it reaches one, or until it has stopped changing for several waiting times. The other, when that is too costly, is to treat the time to the last fusion as the quantity of interest and report its distribution over replicas, rather than a morphology at a fixed time. Larger systems, in which many droplets average out, and enhanced-sampling methods aimed at the distribution of cluster sizes are alternatives.

Many simulation studies of assembling peptides and amphiphiles compare the largest cluster, the number of aggregates or the buried SASA between sequences or conditions, often with one trajectory of fixed length per condition. Our results suggest that such differences can arise purely from when each run happened to be stopped, even for hexane, and that buried SASA computed without periodic images adds variability of its own. For larger molecules with slower internal dynamics, aggregates diffuse more slowly and may merge less readily on contact, so the waiting times may be longer still, although we have not tested this. Robust comparisons can rest on the quantities that do reproduce. These are local structure, conformation, contacts per molecule, the periodically corrected buried surface, early clustering milestones and

energetics. Morphological quantities can be used if each run has demonstrably reached its final state, or if they come with replicas and between-run errors.

The two setup artefacts are invisible within a single run. The concentration shift is a reporting issue and is easily fixed by computing concentrations from the production volume. The SASA artefact is subtler. Its size depends on where a droplet sits, which changes over time and differs between runs. Here it reached 87 nm^2 in individual frames and accounted for about two-thirds of the apparent spread in burial between runs 1–3 at 50 ns; across all ten runs, whose droplets genuinely differed at that time, it added less. At 150 ns, when all ten runs held the same single droplet, it still made their buried shares differ by up to 12.7 percentage points, against 1.4 with periodic images. It is largest exactly when clusters become comparable in size to the box, the regime that assembly studies care about. Recentring the coordinates on a cluster removes it only when there is a single cluster, whereas including periodic images works in general.

In this model, assembly at 310 K raises the enthalpy and is driven entirely by entropy, whereas liquid-hydrocarbon transfer data imply a slightly negative enthalpy at this temperature[34,35]. The decomposition shows that the cost comes from hexane–water dispersion. For each nm^2 of hexane surface buried, about 19 kJ mol^{-1} of hexane–water interaction is lost; hexane–hexane contacts return about 8 kJ mol^{-1} and water and ions about 5 kJ mol^{-1}. The control simulations place the discrepancy in the force field. Relative to experiment, it makes the transfer of hexane from its liquid into water about 7 kJ mol^{-1} too favourable at 25 °C. Half of this comes from weak hexane–hexane cohesion, which the authors of these parameters report for short alkanes[47], and half from hexane–water attraction, set by the combining rules, that is too strong. A cross term fitted to hydration data would be needed before the enthalpy of assembly in such a model can be compared with experiment.

The integration time step, by contrast, is not responsible. Asthagiri recently showed that, with rigid water, time steps above 0.5 fs break the equipartition of kinetic energy between translational and rotational motion[32]. In that work a 4-fs step deepened the methane–methane contact minimum by about 0.25 $k_B T$ relative to 0.5 fs, corresponding to roughly 28% more association. Our 4-fs runs show integration error of this kind: the kinetic temperature runs 1.3 K above its target, and the excess falls to 0.4 K at 2 fs, roughly as the square of the step[33] (Extended Data Fig. 6c). Yet at 2 fs, with or without hydrogen-mass repartitioning, the potential-energy rise and the end state were unchanged, and the density and enthalpy of vaporisation of liquid hexane changed by at most 1.2% (Table 3). The two 2-fs runs completed their last fusion at 71 and 28 ns, within or just below the range of the 4-fs runs; one run per condition cannot detect a modest change in timing, and we did not test steps below 2 fs. The temperature dependence of the water model remains untested. None of these factors bears on the main finding, which arises from the stochastic timing of coalescence rather than from the strength of the attraction.

The study has clear limits. We ran ten replicas of 150 ns at 4 fs and one run at each of two 2-fs settings, at one concentration about 1,950-fold above solubility, with one force field and one water model. Ten runs give a first estimate of the waiting-time distribution, but its tail is poorly constrained: the slowest run took 118 ns, and ten samples cannot say how often a 150-ns run would still hold two droplets. At lower supersaturation, nucleation would be slower and rarer, and we would expect variability between runs to increase. The force field makes hexane less cohesive and more strongly hydrated than experiment, so the energetics reported here are those of the model. The hydration enthalpy was computed only at 4 fs, and its uncertainty of about 1.4 kJ mol^{-1} is too large to resolve the heat-capacity change. Figs 3–5 and the decomposition of the potential energy use runs 1–3 only; the structural, conformational, mobility and surface-area statistics in Table 2 cover all ten runs. The hexane diffusion coefficients are not corrected for finite size, and the droplet diffusion coefficients are upper estimates. Hexane was chosen because it has no specific interactions or slow internal motions. Whether the same split between reproducible and irreproducible quantities, and the same long waits for the last fusion, hold for peptides and amphiphiles remains to be tested.

Conclusions

Simulations of hydrophobic assembly can reproduce how molecules pack, how fast they cluster and, given enough time, what they assemble into, but a single trajectory does not determine when it gets there. For 100 hexanes, ten independent runs that differed in morphology at 50 ns, with a spread in the largest cluster 66 times larger than any one run suggested, all ended as the same single droplet. The final droplet was complete at times from 33 to 118 ns, half the runs had not reached it by 50 ns, and before the last fusion two droplets coexisted for up to 83 ns. Two routine setup steps added further bias. Equilibration changes the concentration, and surface areas computed without periodic images depend on

where a droplet sits. The sign of the assembly enthalpy, finally, was set by the force field's balance of hexane–hexane and hexane–water attraction, not by the time step. We suggest that simulation studies of assembly:

1. run replicas from independent builds, enough to estimate the spread in the time to the end state (here ten), and report the result of each;
2. compare the spread between runs with the error that each run reports;
3. follow the number of droplets or aggregates through time, and report a morphology only once it has stopped changing in every replica, or report the time taken to reach it;
4. give concentrations from the production volume, not the build;
5. compute surface areas with periodic images and whole molecules;
6. compare the first production frame with random placement at the same volume;
7. check the enthalpy of vaporisation and the hydration enthalpy of a single molecule against experiment before interpreting the enthalpy or entropy of assembly;
8. report droplet fusions and other rare events alongside time-averaged results.

These checks cost little. Without them, a trajectory that looks converged can report a state that an independent replica, or the same run continued, would not reproduce.

Methods

System construction

Systems were built with a custom script (build_hexane.py) using OpenMM 8.4.0 (ref. 29). One all-trans n-hexane was generated with the Lipid21 equilibrium geometry: C–C 0.1535 nm, C–H 0.1092 nm, C–C–C 112.05°, H–C–H 108.35° and H–C–C 110.05°. One hundred copies were placed at random positions and orientations in a 9.40-nm cubic box by rejection sampling. No two atoms of different molecules were allowed within 0.5 nm, and no atom within 0.3 nm of a box face. Each hexane was a separate chain (residue HEX). OpenMM's Modeller added OPC water[24] to fill the box, together with Na^+ and Cl^- at 150 mM ionic strength plus neutralising ions (68 of each). Three builds used random seeds 2024, 2025 and 2026, giving 24,929, 24,896 and 24,891 waters and 101,852, 101,720 and 101,700 atoms.

Force field

Hexane used the Amber Lipid21 acyl-chain parameters[25] (atom types cD for carbon and hL for hydrogen), written as an OpenMM residue template (hexane.xml) and combined with amber19/lipid21.xml and amber19/opc.xml. Partial charges are -0.077427 e (C) and +0.025809 e (H) for CH_3, and -0.022816 e (C) and +0.011408 e (H) for CH_2, so every group is neutral. Lennard-Jones parameters are $\sigma = 0.339967$ nm and $\varepsilon = 0.45773$ kJ mol^{-1} for carbon, and $\sigma = 0.260142$ nm and $\varepsilon = 0.04184$ kJ mol^{-1} for hydrogen. Water was the rigid four-site OPC model[24], and ions used the OPC-matched monovalent parameters of Sengupta et al.[26] distributed with amber19/opc.xml. Cross terms followed the Lorentz–Berthelot rules. For example, hexane carbon and water oxygen interact with $\sigma = 0.328$ nm and $\varepsilon = 0.638$ kJ mol^{-1}. One–4 interactions were scaled by the Amber factors (1/1.2 for electrostatics and 1/2 for Lennard-Jones).

The force field was checked against liquid hexane and against the hydration of a single hexane (liquid_hexane.py and hydration_enthalpy.py, run with 11_liquid_hexane.sbatch and 12_hydration_enthalpy.sbatch). For the liquid, 150 hexanes were placed at random in a 4.6-nm cube with a 0.2-nm minimum separation, compressed at 1,000 bar for 0.5 ns, equilibrated at 1 bar for 2 ns and simulated for 20 ns with the production settings. This was done at 298.15 and 310 K, at 4 fs with hydrogen-mass repartitioning and at 2 fs with standard hydrogen masses. A single hexane in vacuum (no cutoff, same integrator, time step and masses) was simulated for 20 ns after 1 ns of equilibration. With N = 150, the enthalpy of vaporisation is

$$\Delta H_{\mathrm{vap}} = \langle U_{\mathrm{gas}} \rangle - \langle U_{\mathrm{liq}} \rangle / N + RT - p \langle V_{\mathrm{liq}} \rangle / N$$

For the hydration enthalpy, one hexane was placed at the centre of a 3.2-nm cube and solvated with 1,065 OPC waters, without ions; deleting the hexane from that box gave a reference with identical waters. Both boxes were equilibrated with the production protocol and simulated for 200 ns at 4 fs, at 298.15 and 310 K, with energies and volumes logged every 1 ps. After the first 1 ns of each run was discarded,

$$\Delta H_{\mathrm{hyd}} = \langle U_{\mathrm{sol}} \rangle - \langle U_{\mathrm{wat}} \rangle - \langle U_{\mathrm{gas}} \rangle + p\,(\langle V_{\mathrm{sol}} \rangle - \langle V_{\mathrm{wat}} \rangle) - RT$$

where −RT removes the pV of the ideal-gas molecule and the kinetic energies cancel. The liquid-to-water transfer enthalpy is the sum of the hydration enthalpy and the enthalpy of vaporisation. Standard errors come from 10 blocks for the liquid and from 20 blocks of 10 ns for the hydration boxes. For the hydration enthalpy they changed by less than 0.5 kJ mol^{-1} for blocks of 1 to 40 ns, against an autocorrelation time of the potential energy of about 2 ps.

Simulation protocol

All simulations used the OpenMM CUDA platform in mixed precision on single NVIDIA A100 GPUs (Theia cluster, University of South Carolina). Electrostatics used particle-mesh Ewald[37,38] with a 1.0-nm real-space cutoff and OpenMM's default error tolerance (5×10^{-4}). Lennard-Jones interactions were truncated at 1.0 nm with the long-range dispersion correction. Bonds to hydrogen were constrained, and water was kept rigid[39,40]. Hexane hydrogens were given a mass of 1.5 amu, taken from the bonded carbon[30,31]; water hydrogens were not modified. This allowed a 4-fs time step. Temperature was held at 310 K with the LangevinMiddle integrator[41] (friction 1 ps^{-1}), and pressure at 1 bar with a Monte Carlo barostat attempting volume moves every 25 steps[42,43].

Equilibration (run_md.py) proceeded in four stages. (0) Energy was minimised for up to 30,000 iterations, with harmonic restraints on hexane carbons (k = 1,000 kJ mol^{-1} nm^{-2}). (1) The system was simulated at constant volume for 10 ps with a 0.5-fs step, starting from velocities drawn at 50 K, and then heated from 50 to 310 K in ten steps over 100 ps at 4 fs, with restraints maintained. (2) Under constant pressure, restraints were released over five 100-ps stages (k = 1,000, 500, 200, 50 and 0 kJ mol^{-1} nm^{-2}). (3) Finally, the system was simulated without restraints for 0.1 ns at constant pressure. This last stage was kept short because hexane at this concentration begins clustering within about a nanosecond, and clustering during equilibration would not be recorded. Production ran for 150 ns per replica. Hexane coordinates were saved every 10 ps (15,000 frames per run), full-system coordinates every 500 ps (300 frames per run), energies and box volume every 10 ps, and checkpoints every 200 ps. Runs were restartable from checkpoints, and segments were concatenated with overlapping frames removed. Each run was first taken to 50 ns (run 1 in stages, from 10 to 15, 20 and 50 ns) and then extended to 150 ns from its 50-ns checkpoint with unchanged settings (06_extend.sbatch). Across the 50-ns restart, the mean atomic displacement between consecutive 10-ps hexane frames (0.37–0.46 nm) matched that between neighbouring frames, so the trajectories are continuous. Replica k used velocity seed 1000k + 29, integrator seed 1000k + 13 and barostat seed 1000k + 7, so no random input was shared between replicas (Table 1). Throughput was 343–351 ns per day.

To observe the restrained stages, the equilibration of each of the three boxes was repeated with the original seeds and with volume, density, temperature and energy logged every 1 ps through all stages (`run_md.py --log-equil`, 13_equil_logged.sbatch). The repeats are statistically equivalent to the original equilibrations but not identical, because GPU arithmetic is not bitwise reproducible between runs; their final volumes differ from the originals by 0.3–2.6 nm^3, within the fluctuation of the box. For the time-step control (10_dt2_control.sbatch), run 1 was repeated from the same starting box and seeds at a 2-fs step, once with hydrogen-mass repartitioning and once with standard hydrogen masses. For the latter the box was rebuilt with build seed 2024; the builder is deterministic, and the rebuilt box matches run 1's in packing attempts (126), water molecules (24,929) and ions. Both runs used the same equilibration and 150 ns of production, at 177 and 168 ns per day. Clusters, fusions and milestones were computed as for runs 1–3 (hex_dt2.py), which reproduces the values reported for runs 1–3 exactly.

Replicas 4–10 (09_new_replicas.sbatch) were built and run exactly as runs 1–3, with build seeds 2028–2034 and replica k using velocity seed 1000k + 29, integrator seed 1000k + 13 and barostat seed 1000k + 7. Their boxes held 24,890–24,929 waters, and each reached 150 ns in a single 24-h allocation at 339–355 ns per day. They were analysed with the definitions used for runs 1–3 (hex_ten.py), which reproduces the values reported for runs 1–3 exactly. Milestones, fusions and the coexistence time of the last two droplets were computed as in the following sections. Periodic surface areas, radial distribution functions, contacts, dihedral conformations, mobility and droplet tracking were computed for runs 4–10 with the same scripts as for runs 1–3 (run_all10.sh). The statistics in Table 2, and the panels of Figs 1, 2 and 6 that show all ten runs, were computed with stats10.py and paper_figs_v5.py. The energy decomposition was computed for runs 1–3 only.

Concentrations were computed from the mean production volume $\langle V \rangle$:

$$c = \frac{N}{N_A \langle V \rangle}, \qquad N = 100$$

Cluster analysis

Two hexanes were in contact if any pair of their carbon atoms lay within 0.5 nm (minimum-image convention). Clusters were the connected components of the contact graph, found with SciPy[44]. An independent recomputation from the trajectories reproduced the original cluster sizes in all 45,000 frames. For every frame we recorded the number of clusters, the size of the largest cluster, and the number of molecules in clusters of two or more. Free molecules were those with no contact. Time series were smoothed with a centred 0.5-ns (50-frame) running mean. Milestone times are the first time after which the smoothed series never again falls below a threshold: a largest cluster of 10 (stable nucleus), 30, 50 or 90 molecules, or 90 molecules in clusters. Contacts per hexane is the number of other hexanes in contact with each molecule, averaged over molecules and frames.

Droplets, fusions and transfers

Droplets are clusters of at least ten molecules (hex_exchange.py). Each droplet was followed from frame to frame by membership. A droplet inherits the identity of the previous-frame droplet with which it shares the most molecules, provided they share more than half of the smaller one; otherwise it receives a new identity. A fusion (hex_fusions_all.py) is a frame in which one cluster contains at least five molecules from each of two or more clusters of the previous frame that each had at least five members. In addition, at least 90% of the merged molecules must stay in one cluster at every frame of a 0.2-ns window. The first-contact time is the frame of the merger, and the hold time is the start of the first such window. A transfer is a molecule whose most recent droplet changes from one droplet to another while both exist, so that joining a droplet by fusion is not counted. For the last fusion of each run (hex_lastpair.py), the two parent droplets were followed back in time by membership for as long as both existed, and their closest carbon–carbon distance (minimum image) was recorded every 10 ps. An approach episode begins when that distance falls below 1.0 nm and ends when it rises above 1.5 nm (hex_waiting.py). Droplet centres are the means of the members' centres of mass, unwrapped across periodic boundaries and followed continuously in time. Droplet diffusion coefficients are linear fits of the centre MSD over lags of 0.1–2 ns, for every droplet followed without interruption for at least 5 ns; they are upper estimates, because molecules joining or leaving a droplet also move its centre. For the single droplet that remained after the last fusion, followed from 1 ns after the fusion to 150 ns, the fit used lags of 1–10 ns. The Smoluchowski waiting time is 1/k with k $= 4\pi(D_A + D_B)(R_A + R_B)/\langle V \rangle$[12], using the mean diffusion coefficient of the droplets followed before their last fusion for both droplets. The radii are those of spheres of liquid hexane in this model (0.2254 nm^3 per molecule) holding each droplet's mean number of molecules, with the contact distance taken as $R_A + R_B$ or $R_A + R_B + 0.5$ nm. The dihedral angles C1–C2–C3–C4, C2–C3–C4–C5 and C3–C4–C5–C6 were evaluated every 0.1 ns (hex_gauche.py), and a dihedral counts as gauche when $|\phi| < 120°$.

Molecular renderings

Snapshots were drawn with a custom z-buffered spacefill renderer (hex_render.py) in orthographic projection, with radii of 0.17 nm for carbon and 0.11 nm for hydrogen. Shading combines diffuse and specular lighting, screen-space ambient occlusion, depth fog and silhouette outlines. Each frame was first shifted so that the largest cluster sat at the box centre; for fusion sequences, the midpoint of the two merging droplets was used instead. Molecules were then placed in the box by their centres of mass, with each cluster drawn whole around its own centre. In Figs 1a and 2a,b, droplets are coloured by size rank in that frame. In the fusion sequences (Extended Data Fig. 2), each molecule keeps the colour of the droplet it belonged to in the first frame of the sequence.

Random-placement reference

The 100 hexanes were placed at random in a cube of 773.1 nm^3, keeping the internal geometry of the starting conformers, with uniformly random centres and orientations (30 trials per variant, random_baseline_trials.py). In the ideal variant,

overlaps were allowed. In the hard-core variant, molecules were inserted sequentially, and any placement that brought heavy atoms of different molecules within 0.3 nm was rejected. Cluster statistics used the same 0.5-nm criterion as above.

Surface area

Solvent-accessible surface area was computed every 0.1 ns with the Shrake–Rupley algorithm[22] as implemented in MDTraj[23]. We used MDTraj's default atomic radii and 960 sphere points per atom, with a 0.14-nm probe, on whole molecules and without periodic images. The isolated-molecule reference A_i^{iso} is the SASA of each hexane computed alone in its conformation in the same frame; it averaged 3.03 nm^2. The buried share of hexane surface is

$$f_{\mathrm{bur}}(t) = \frac{\sum_{i=1}^{N} A_i^{\mathrm{iso}}(t) - A_{\mathrm{tot}}(t)}{\sum_{i=1}^{N} A_i^{\mathrm{iso}}(t)}$$

where A_{tot} is the SASA of all hexanes together. Periodic SASA (sasa_pbc.py) wraps every atom into the box. It then appends the atoms of the 26 neighbouring images that lie within 0.8 nm of a face, computes Shrake–Rupley SASA for the combined set with the same radii, probe and sphere points, and sums only the central atoms. The buffer of 0.8 nm exceeds twice the largest atomic radius plus probe radius, so every atom sees all of its true neighbours. For the droplet test (hex_droplet_pbc.py), 95 hexanes were inserted at random into a sphere, with a carbon–carbon hard core of 0.37 nm. The sphere radius was increased from 1.6 nm until all molecules fitted, which happened at 1.8 nm, with all atoms within 2.3 nm of the centre. The droplet was translated to the centre, a face, an edge and a corner of a 9.18-nm box. Buried shares were computed with molecules kept whole and wrapped by centre (uncorrected), and with periodic images. For the three runs, periodic and uncorrected SASA were computed from the same frames every 0.1 ns (hex_sasa_pbc.py). The uncorrected values reproduce the standard analysis (analyze.py) to within 0.3 nm^2 wherever both were computed (runs 1 and 2 over 150 ns, run 3 over its first 50 ns), and the periodic result does not change when the whole box is shifted or the 0.8-nm buffer is enlarged. Surface areas in the Results are periodic unless labelled uncorrected.

Structure

The centre-of-mass RDF was computed from all hexane pairs in every frame under the minimum-image convention, in 60 bins up to half the shortest box edge (bin width 0.077 nm). It was normalised by the ideal-gas pair density at the mean box volume:

$$g(r) = \frac{\langle n(r) \rangle}{\frac{1}{2} N(N-1)\, \Delta V(r) / \langle V \rangle}$$

$$w(r) = -k_{\mathrm{B}} T \ln g(r)$$

Here n(r) is the number of pairs in the shell at r, ΔV(r) the shell volume, and $k_{\mathrm{B}}T$ = 2.578 kJ mol^{-1} at 310 K. The first minimum r_{min} is the first local minimum after the main peak. The coordination number is

$$n_1 = \frac{N-1}{\langle V \rangle} \int_0^{r_{\mathrm{min}}} 4\pi r^2 g(r)\, \mathrm{d}r$$

The barrier in w(r) is w(r_{min}) minus the well minimum.

Mobility

Centre-of-mass trajectories (every 50 ps) were unwrapped by accumulating minimum-image displacements between frames (mobility.py). The MSD was averaged over molecules and time origins:

$$\mathrm{MSD}(\tau) = \left\langle |\mathbf{r}_i(t+\tau) - \mathbf{r}_i(t)|^2 \right\rangle_{i,t} \approx 6D\tau + c$$

D was obtained from a linear fit over τ = 1–10 ns, for the whole run (mobility.py) and separately for 0–50, 50–100 and 100–150 ns (hex_mobility_windows.py), with time origins restricted to each window. The apparent diffusion coefficient is MSD(τ)/6τ. No finite-size correction[36] was applied to hexane diffusion. The net displacement is the distance between

each molecule's unwrapped first and last positions. Partners are molecules that came within 1.5 nm (centre of mass) at least once. Rotation was measured by the autocorrelation of the unit vector from C1 to C6, and its 1/e time is reported to the 50-ps resolution of the sampling. The Stokes–Einstein estimate for a droplet is $D = k_B T/(6\pi\eta R)$ with stick boundary conditions, and the finite-size correction for a cubic box of edge L is $k_B T\xi/(6\pi\eta L)$ with $\xi = 2.837297$ (ref. 36).

Energetics

Potential energy and box volume were recorded every 10 ps. The rise in potential energy is the mean over 145–150 ns minus the mean over 0–5 ns, and for comparison the mean over 45–50 ns minus the mean over 0–5 ns. Its uncertainty combines the standard errors of the two means, each from five 1-ns blocks. To relate potential energy to burial, 1-ns block means of potential energy (relative to the 0–5 ns mean) were paired with 1-ns block means of the periodic buried surface. Pearson correlations and least-squares slopes were computed for each run and for all 450 blocks pooled. The pressure–volume term was estimated as $p\Delta V$ at 1 bar.

The potential energy was decomposed exactly (hex_energy_split.py) on 205 saved full-system frames of each run: every frame (0.5 ns apart) up to 50 ns, one frame per nanosecond from 51 to 145 ns and every frame over 145.5–150 ns. The system was rebuilt in OpenMM 8.6 from the final structure of each run and the same force-field files, and each frame was evaluated on the CPU platform. The recomputed potential energies agreed with the logged ones to within 1.8 kJ mol^{-1}. Atoms were divided into three groups: hexane (H), water including the OPC virtual sites (W) and ions (I). A copy of the nonbonded force, without the long-range dispersion correction, was given parameter offsets that switch each group's charges, or its Lennard-Jones well depths, off. The PME energy is a quadratic form in the charges, and the Lennard-Jones energy is a sum over pairs. The interaction between groups X and Y is therefore exactly E(X+Y) − E(X) − E(Y), evaluated separately for electrostatics and Lennard-Jones; the pieces sum to the total nonbonded energy to within 0.01 kJ mol^{-1}. The dispersion correction depends only on the box volume here and is assigned to the water-and-ion term. Hexane intramolecular nonbonded energy (scaled 1–4 pairs and more distant pairs within a molecule, plain Coulomb and Lennard-Jones within 1.0 nm) was computed directly and removed from the hexane–hexane term. Together with the hexane bond, angle and torsion energies, it is reported as hexane internal energy. Water–water and water–ion energies fluctuate by about 10^3 kJ mol^{-1} between frames. Their change was therefore also obtained by difference: the logged change in potential energy (500 samples per window) minus the change in all hexane terms (10 frames per window, with small scatter). Uncertainties of window differences are standard errors combined in quadrature.

For the water-and-ion control (07_water_control.sbatch), a box with no hexane was built with the same builder and settings (build seed 2027). It contains 27,033 OPC waters and 74 Na^+ and 74 Cl^- ions, 108,280 atoms in all. Only amber19/opc.xml is loaded for this box; it supplies the same water and ion parameters as in the hexane systems, which we confirmed parameter by parameter. The box was equilibrated and simulated for 150 ns with the same protocol as the hexane runs (50 ns, then extended with 08_extend_water_control.sbatch). Its change in potential energy between 0–5 ns and 45–50, 95–100 and 145–150 ns, and its linear drift over 1-ns blocks across the whole run, were computed with control_check.py. In Extended Data Fig. 1d its potential energy is scaled by the ratio of water molecules, 24,905/27,033.

Reproducibility statistics

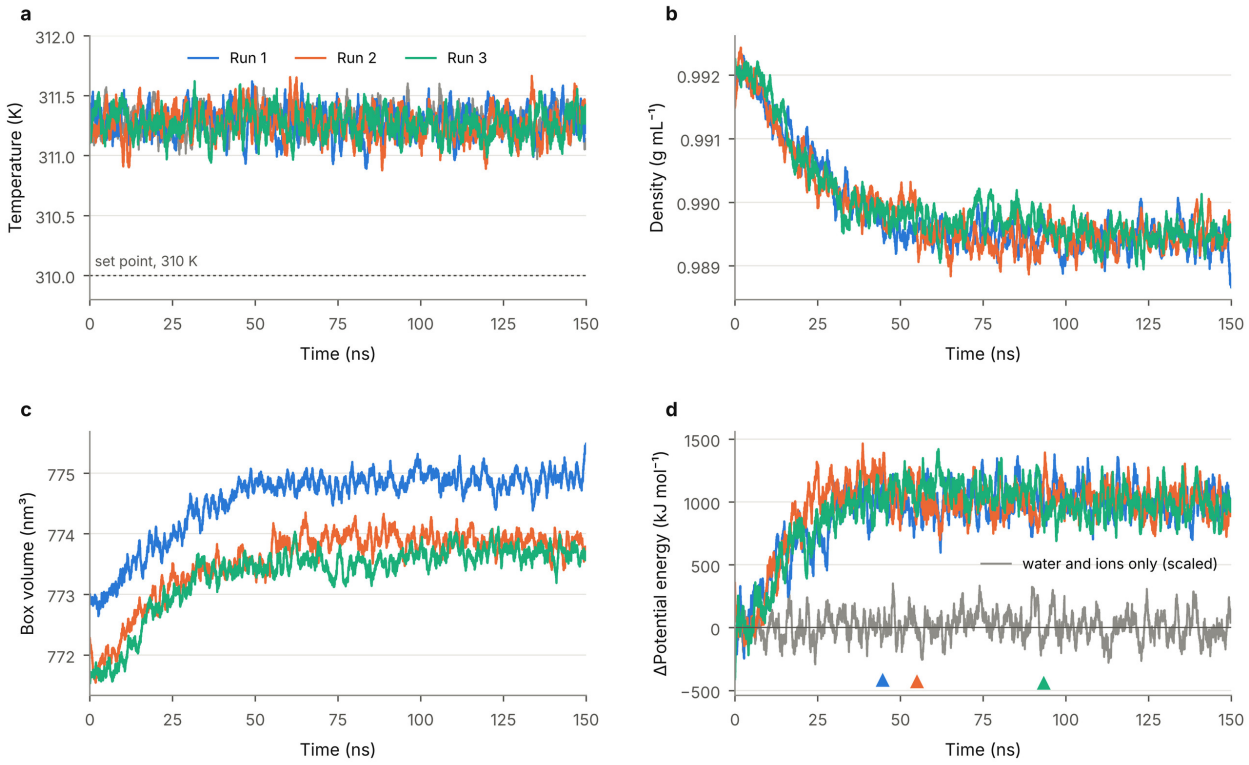
For each quantity x, each run k and each 5-ns window, the window value $\bar{x}_k$ is the mean over the window, and its standard error is computed from five 1-ns block means $b_{k,j}$ (hex150_stats.py):

$$\mathrm{SEM}_k = \frac{s\left(\{b_{k,j}\}_{j=1}^{5}\right)}{\sqrt{5}}$$

$$s_{\mathrm{between}} = \sqrt{\frac{1}{n-1}\sum_{k=1}^{n}(\bar{x}_k - \bar{x})^2}$$

$$R = \frac{s_{\mathrm{between}}}{\frac{1}{n}\sum_k \mathrm{SEM}_k}$$

Here n = 3 for runs 1–3 and n = 10 for all ten runs. R is the reproducibility ratio shown in Fig. 2d, and Fig. 2e shows its numerator and denominator for the largest cluster; R is near 1 when a single run estimates its own uncertainty correctly.



Extended Data Fig. 1 | Thermodynamic stability of the three runs. a–d, Temperature, density, box volume and potential energy (relative to the 0–5 ns mean) against time, as 1-ns running means of values logged every 10 ps. Temperature stays about 1.3 K above the 310-K set point throughout. As hexane assembles, the density falls, the box volume grows by about 2 nm^3 and the potential energy rises (Fig. 3); all three level off after the last droplet fusion of each run (triangles in **d**). Run 1 contains 33–38 more water molecules than runs 2 and 3, hence its larger volume. Grey lines are the hexane-free box of water and ions, run for 150 ns with the same protocol; in d its potential energy is scaled to the mean water content of the hexane runs, and it shows no drift (-0.01 ± 0.22 kJ mol^{-1} per ns).

It was computed for 45–50, 95–100 and 145–150 ns and for every 5-ns window from 0–5 to 145–150 ns; it is undefined when a quantity does not vary at all within or between runs. For the potential-energy rise, the single-run error combines the standard errors of the 0–5 and 145–150 ns windows. The coefficient of variation is $s_{\text{between}}/\bar{x}$. Other whole-run quantities (RDF features, contacts, coordination, D and milestones) were compared through their CV only. Values in the text are given as mean ± s.d. over runs 1–3 or over all ten runs, as stated; Table 2 gives both. The 95% confidence interval for a standard deviation estimated from three values is [0.52s, 6.3s] (χ^2 distribution, two degrees of freedom), and from ten values [0.69s, 1.83s] (nine degrees of freedom).

Software

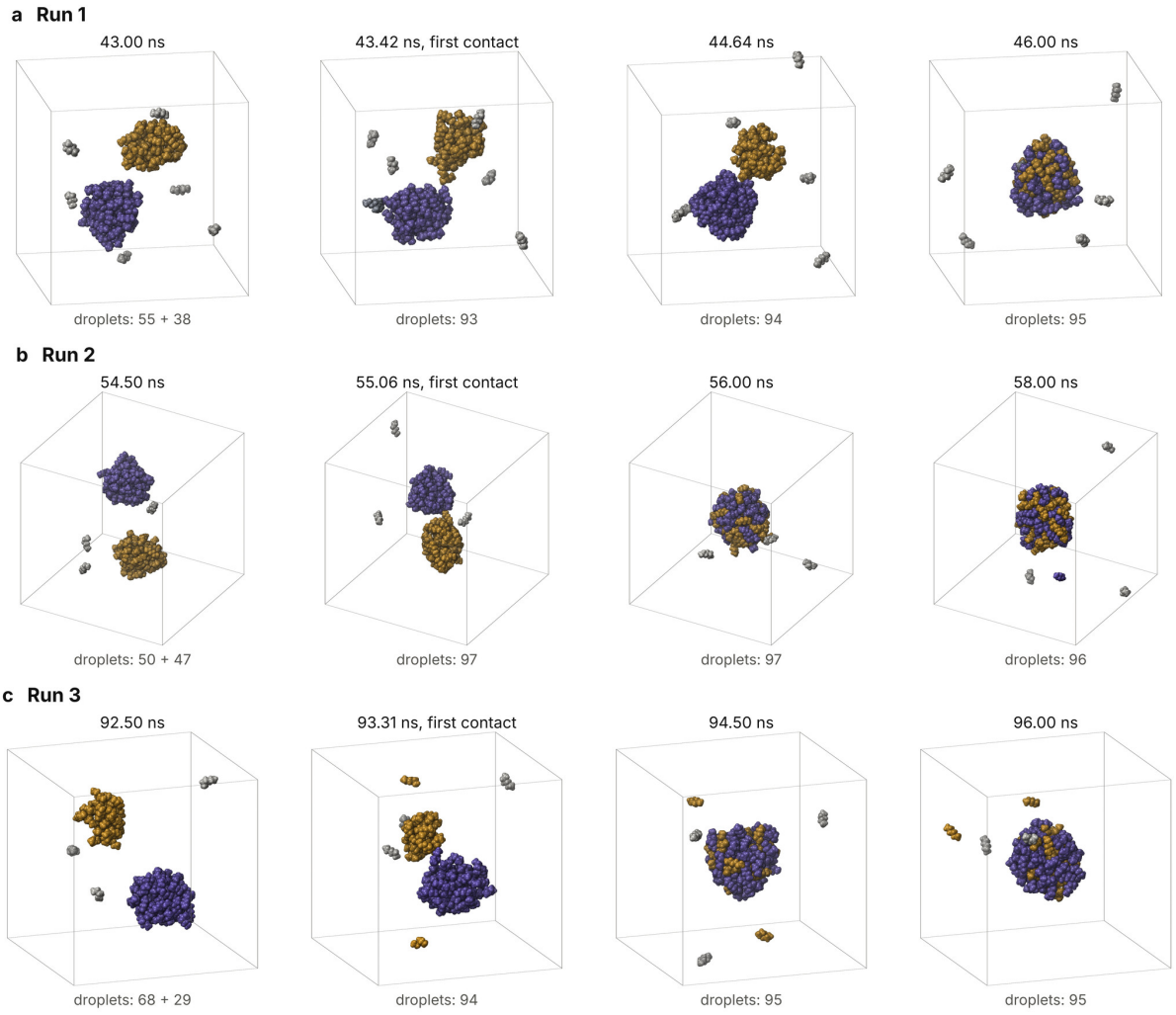
Simulations used OpenMM 8.4.0 (ref. 29). Analysis used MDTraj[23], NumPy[45], SciPy[44] and pandas, and figures were drawn with Matplotlib[46].

Use of AI tools

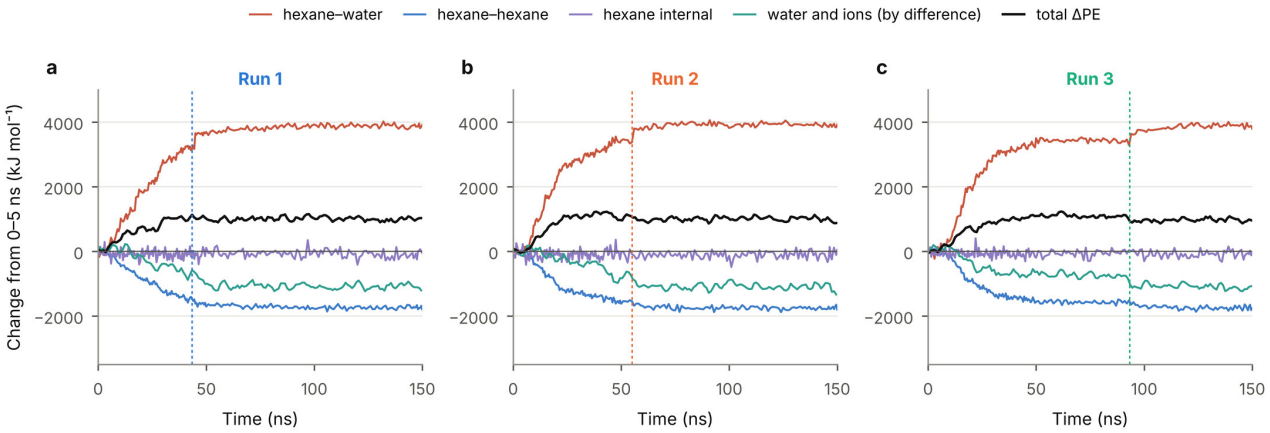
Simulation and analysis scripts and a first draft of this manuscript were prepared with the assistance of an AI model (Claude, Anthropic). All simulations were run by the authors, and all numbers were computed from the simulation outputs by the scripts listed under Code availability. The authors checked the text, values and references against those outputs and the cited sources, and take full responsibility for the content.

Extended Data

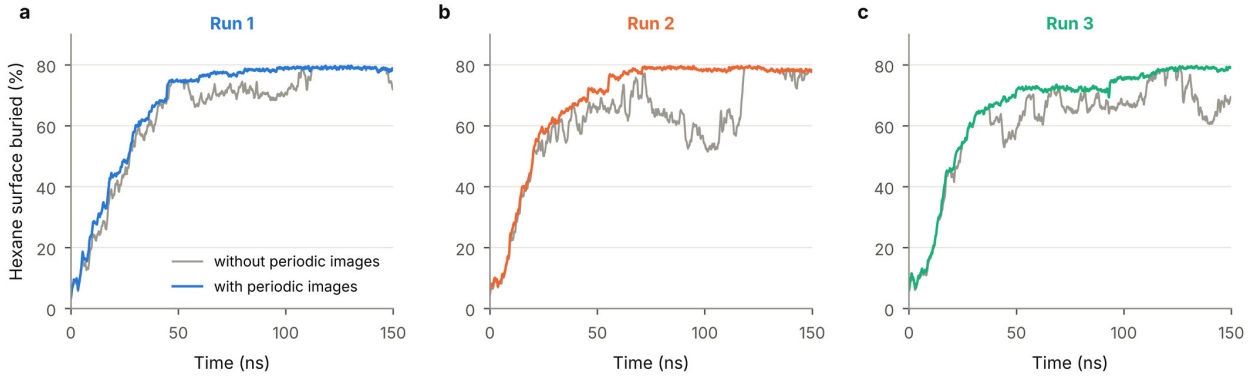
Figures 1–6 appear with their legends in the Results. The Extended Data figures below support them.



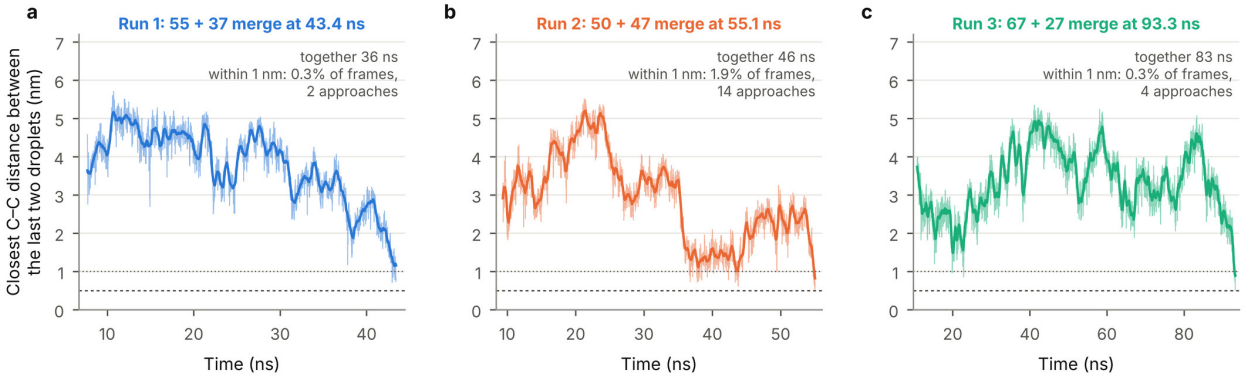
Extended Data Fig. 2 | The last droplet fusion of each run. a, Run 1 at 43.00 ns, 43.42 ns (first contact), 44.64 ns and 46.00 ns. **b**, Run 2 at 54.50 ns, 55.06 ns (first contact), 56.00 ns and 58.00 ns. **c**, Run 3 at 92.50 ns, 93.31 ns (first contact), 94.50 ns and 96.00 ns. Molecules keep the colour of the droplet they belonged to in the first frame of each row: purple for the larger and gold for the smaller of the two merging droplets (55 and 38 molecules in **a**, 50 and 47 in **b**, 68 and 29 in **c**). Grey marks all other molecules. Each row is centred on the two merging droplets and drawn from one fixed direction, and the numbers give the droplet sizes (at least ten molecules) in each frame. Within 1–3 ns of first contact, the molecules of the two droplets are intermixed.



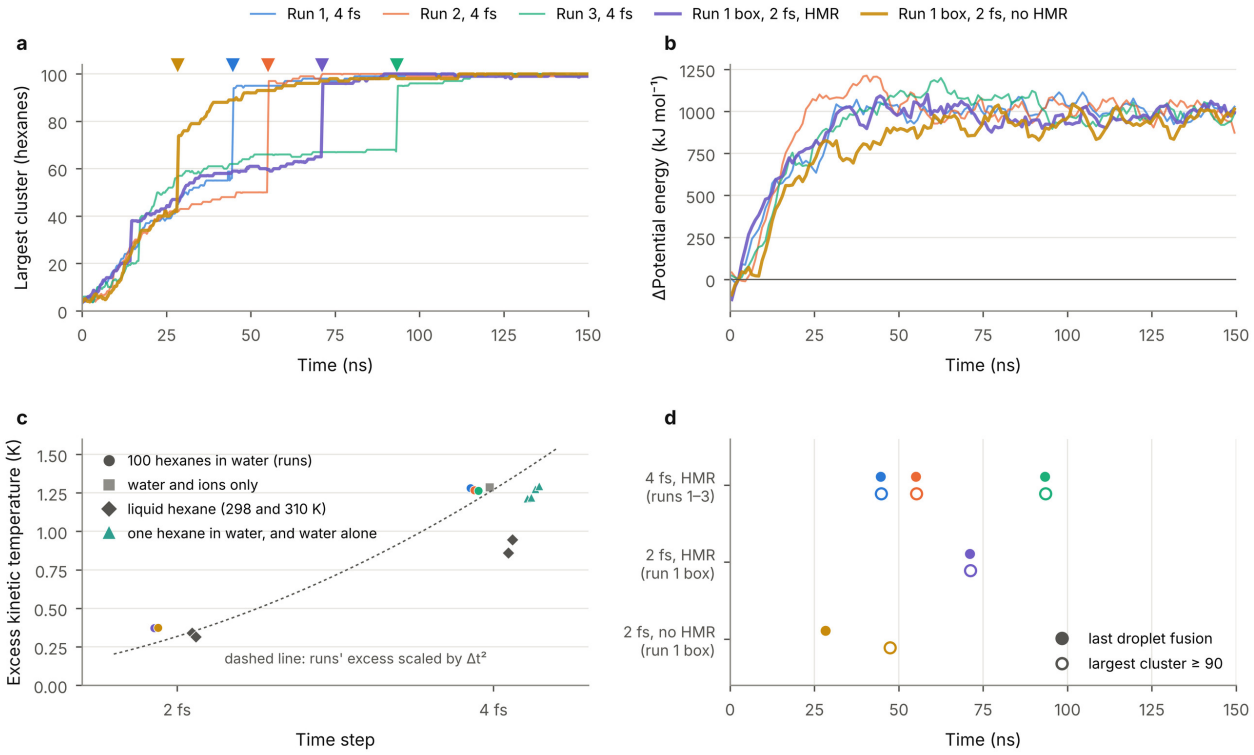
Extended Data Fig. 3 | Energy decomposition over time. a–c, Runs 1–3. Each panel shows the change from the 0–5 ns mean in the hexane–water, hexane–hexane and hexane internal energies at each analysed full-system frame (every 0.5 ns to 50 ns, every 1 ns from 51 to 145 ns and every 0.5 ns over 145.5–150 ns). It also shows the total potential energy and the water-and-ion term, obtained as the total minus all hexane terms (1-ns means of values logged every 10 ps, 3-ns running mean). Dashed lines mark the last droplet fusion of each run.



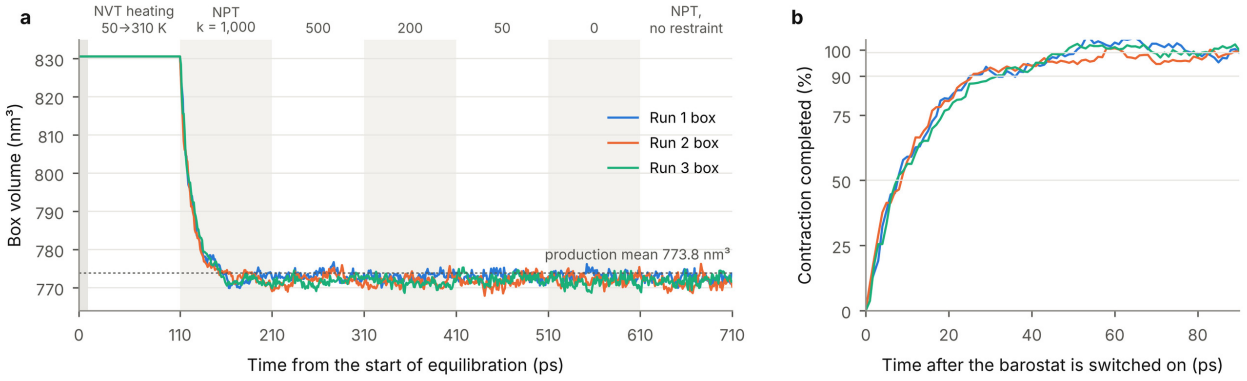
Extended Data Fig. 4 | Surface burial with and without periodic images. Buried share of hexane surface in each run against time, computed every 0.1 ns without (grey) and with (colour) periodic images, shown as 0.5-ns running means.



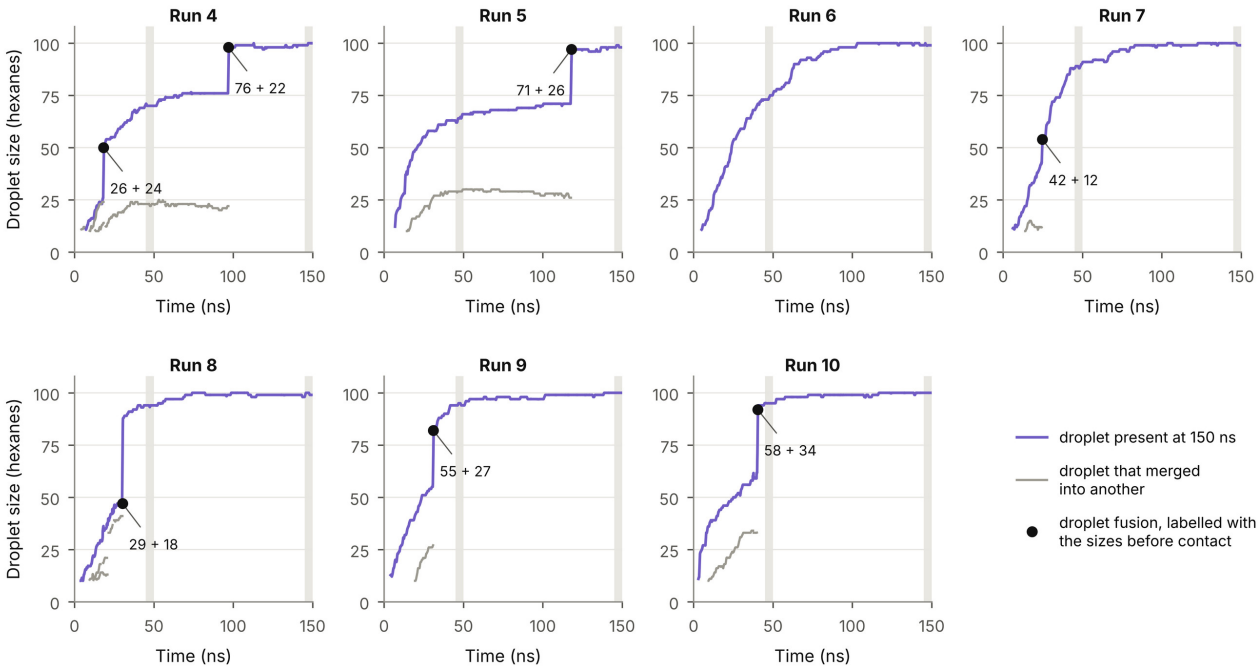
Extended Data Fig. 5 | The last two droplets of each run before they merged. a–c, Closest carbon–carbon distance between the two droplets whose fusion ended each run (minimum image), every 10 ps (thin) and as a 0.5-ns running mean (thick), from the time both droplets first reached ten molecules to their first contact. The dashed line marks the 0.5-nm contact distance and the dotted line 1 nm. The text gives the time the two droplets coexisted, the share of frames in which they were within 1 nm of each other and the number of approach episodes (below 1.0 nm, ending above 1.5 nm).



Extended Data Fig. 6 | Time-step control. Run 1 repeated from the same box and seeds at a 2-fs step, with hydrogen-mass repartitioning (purple) and with standard hydrogen masses (gold), alongside runs 1–3 at 4 fs. **a**, Largest cluster (0.5-ns running mean); triangles mark the last droplet fusion of each run. **b**, Potential energy relative to the 0–5 ns mean (1-ns blocks, 5-ns running mean). **c**, Mean kinetic temperature above the set point for every system run at both steps: the hexane runs, the water-and-ion control, liquid hexane at 298 and 310 K, and the two hydration boxes at each temperature. The dashed line scales the hexane runs' excess by the square of the time step. **d**, Times of the last droplet fusion (filled) and of the first persistent cluster of at least 90 molecules (open). In the run without repartitioning the last fusion of two droplets (40 + 26 molecules) came at 28.3 ns, and the remaining molecules joined as clusters of fewer than ten.



Extended Data Fig. 7 | The box contracts in the first restrained constant-pressure stage. **a**, Box volume, logged every 1 ps, through the 0.71-ns equilibration of each of the three boxes, repeated with the original seeds. Shading separates the stages, and the numbers above them give the restraint constant on hexane carbons (kJ mol^{-1} nm^{-2}). The darker band at 0–10 ps is the warm-up at 0.5 fs. The dashed line is the mean production volume, which also includes about 2 nm^3 of growth during assembly. **b**, Fraction of the contraction completed after the barostat is switched on at 110 ps.



Extended Data Fig. 8 | Droplets in runs 4–10. Size of every droplet (cluster of at least ten molecules) in each of runs 4–10 against time, each droplet followed by its membership (0.5-ns running mean), drawn as in Fig. 2c. Purple lines are the droplet present at 150 ns and grey lines droplets that merged into another. Dots mark fusions between two droplets, at the time each held, labelled with their sizes just before contact. Shaded bands mark 45–50 and 145–150 ns. In run 6 a single droplet grew to its final size by absorbing monomers and small clusters, without a droplet fusion.

Supplementary Table 1 | Lennard-Jones cross terms. Lorentz–Berthelot σ, ε and $r_{\min}$ for every pair of Lennard-Jones sites (hexane, OPC water, Na^+ and Cl^-), from hex_ff_pairs.csv.

Supplementary Data 1 | Per-run values for all ten runs. Every quantity in Table 2 for each of runs 1–10 (per_run_all.csv), the window statistics behind Fig. 2d,e (windows10.csv and sliding10.csv) and every droplet fusion between clusters of at least ten molecules (ten_fusions_min10.csv).

Tables

Table 1 | Systems and replicas.

Quantity	Run 1	Run 2	Run 3	Mean ± s.d.
Build seed	2024	2025	2026	–
Velocity / integrator / barostat seed	1029 / 1013 / 1007	2029 / 2013 / 2007	3029 / 3013 / 3007	–
Hexane molecules	100	100	100	–
Water molecules	24,929	24,896	24,891	–
Na^+ / Cl^-	68 / 68	68 / 68	68 / 68	–
Total atoms	101,852	101,720	101,700	–
Box volume as built (nm^3)	830.6	830.6	830.6	–
Nominal concentration (mM)	199.9	199.9	199.9	–
Production box volume (nm^3)	774.6	773.6	773.3	773.8 ± 0.7
Production box edge (nm)	9.184	9.180	9.179	9.181 ± 0.003
Hexane concentration (mM)	214.4	214.7	214.7	214.6 ± 0.2
Kinetic temperature (K)	311.28	311.27	311.26	311.27 ± 0.01
Density (g mL^{-1})	0.98985	0.98985	0.98995	0.98988 ± 0.00006
Production length (ns) / frames	150 / 15,000	150 / 15,000	150 / 15,000	–
Throughput (ns per day, A100)	342.8	351.0	344.8	346.2 ± 4.3

Temperature, density and volume are means over the full 150 ns of production. The concentration is $100/(N_\mathrm{A}\langle V\rangle)$. Shared settings are given in Methods.

Table 2 | Results for runs 1–3 and for all ten runs.

Quantity	Runs 1–3, mean ± s.d.	CV (%)	R	All ten runs, mean ± s.d.	CV (%)	R
Assembly kinetics						
Hexanes in clusters, first frame	22.0 ± 3.6	16	–	25.7 ± 6.2	24	–
Peak number of clusters ≥2	17.1 ± 1.9	11	–	17.0 ± 1.3	7.7	–
Stable nucleus, largest ≥10 (ns)	6.64 ± 2.11	32	–	5.22 ± 1.89	36	–
Largest ≥30 from (ns)	16.98 ± 0.11	0.62	–	15.04 ± 3.27	22	–
Largest ≥50 from (ns)	36.67 ± 15.25	42	–	27.82 ± 9.92	36	–
Largest ≥90 from (ns)	64.52 ± 25.61	40	–	63.67 ± 29.11	46	–
90 hexanes in clusters from (ns)	35.19 ± 3.34	9.5	–	40.13 ± 9.97	25	–
Last droplet fusion, held from (ns)[b]	64.3 ± 25.6	40	–	59.4 ± 34.4 (n = 9)	58	–
State at 45–50 ns (what a 50-ns study would report)						
Largest cluster	69.3 ± 23.1	33	90	78.8 ± 16.6	21	66
Clusters of ≥2	1.73 ± 0.63	36	12	1.63 ± 0.65	40	8.4
Free hexanes	4.4 ± 1.7	39	7.4	7.7 ± 5.6	73	20
Hexanes in clusters	95.6 ± 1.7	1.8	7.4	92.3 ± 5.6	6.1	20
Hexane SASA (nm^2)	86.0 ± 8.2	9.6	8.4	90.4 ± 15.4	17	19
Hexane SASA, uncorrected (nm^2)[a]	105.2 ± 25.8	25	16	107.9 ± 16.9	16	6.8
Buried share (%)	71.6 ± 2.7	3.8	8.4	70.2 ± 5.1	7.3	19
Buried share, uncorrected (%)[a]	65.3 ± 8.5	13	16	64.4 ± 5.6	8.7	6.8
Density (g mL^{-1})	0.98970 ± 0.00015	0.015	2.3	0.98974 ± 0.00019	0.019	2.3
Temperature (K)	311.35 ± 0.06	0.018	0.94	311.31 ± 0.06	0.02	1.4
End state, 145–150 ns						
Largest cluster	99.3 ± 0.3	0.26	1.8	99.4 ± 0.5	0.5	3.5
Clusters of ≥2	1.00 ± 0.00	0	–	1.00 ± 0.00	0	–
Free hexanes	0.7 ± 0.3	35	1.8	0.6 ± 0.5	78	3.5
Hexanes in clusters	99.3 ± 0.3	0.26	1.8	99.4 ± 0.5	0.5	3.5
Hexane SASA (nm^2)	65.6 ± 0.9	1.4	2.3	65.2 ± 1.3	2	2.9
Hexane SASA, uncorrected (nm^2)[a]	79.7 ± 15.3	19	7.6	79.8 ± 13.2	17	7.1
Buried share (%)	78.4 ± 0.3	0.38	2.3	78.5 ± 0.4	0.54	2.9
Buried share, uncorrected (%)[a]	73.7 ± 5.0	6.8	7.6	73.7 ± 4.4	5.9	7.1
Density (g mL^{-1})	0.98943 ± 0.00012	0.012	1.4	0.98940 ± 0.00009	0.0087	1.2
Temperature (K)	311.33 ± 0.05	0.015	1.0	311.28 ± 0.05	0.016	0.85
Droplets						
Droplets at 50 ns	1.67 ± 0.58	35	–	1.40 ± 0.52	37	–
Last two droplets: time together (ns)[b]	54.6 ± 24.7	45	–	34.1 ± 25.7 (n = 9)	75	–
Last two droplets within 1 nm (% of frames)[b]	0.8 ± 0.9	–	–	2.0 ± 1.6 (n = 9)	–	–
Single droplet diffusion, 1–10 ns (nm^2 ns^{-1})[b]	0.064 ± 0.009	14	–	0.059 ± 0.007 (n = 9)	12	–
Structure and conformation, 0–150 ns						
RDF first peak (nm)	0.5724 ± 0.0002	0.034	–	0.5724 ± 0.0001	0.024	–
RDF peak height	30.99 ± 0.56	1.8	–	30.68 ± 0.99	3.2	–
RDF first minimum (nm)	0.8776 ± 0.0003	0.034	–	0.8777 ± 0.0002	0.024	–
Neighbours in first shell	8.33 ± 0.20	2.4	–	8.28 ± 0.29	3.5	–
Contacts per hexane	6.43 ± 0.14	2.2	–	6.38 ± 0.22	3.4	–
w(r) well (kJ mol^{-1})	−8.85 ± 0.05	0.53	–	−8.82 ± 0.08	0.95	–
w(r) barrier to second shell (kJ mol^{-1})	1.62 ± 0.04	2.7	–	1.61 ± 0.05	3.2	–
Gauche dihedrals, clustered molecules (%)	42.9 ± 0.0	–	–	43.0 ± 0.1	–	–
Gauche dihedrals, free molecules (%)	45.9 ± 0.2	–	–	45.8 ± 0.2	–	–
Energetics, 0–5 → 145–150 ns (kJ mol^{-1})						
Potential-energy rise (logged)	+952 ± 27	2.8	0.39	+940 ± 65	6.9	0.94
Potential-energy rise to 45–50 ns, for comparison	+1,057 ± 33	3.1	–	+924 ± 133	14	–
Hexane–water	+3,880 ± 9	–	–	runs 1–3 only	–	–
Hexane–hexane (intermolecular)	−1,741 ± 3	–	–	runs 1–3 only	–	–
Hexane internal (bonded and intramolecular)	−84 ± 13	–	–	runs 1–3 only	–	–
Hexane–ions	+31 ± 5	–	–	runs 1–3 only	–	–
Water and ions (by difference)	−1,134 ± 10	–	–	runs 1–3 only	–	–
Box volume change (nm^3)	+2.03 ± 0.08	–	–	+1.99 ± 0.08	–	–
Correlation of potential energy with buried area (r)	0.85 ± 0.03	–	–	0.87 ± 0.03	–	–
Rise per buried area (kJ mol^{-1} nm^{-2})	4.1 ± 0.2	–	–	4.1 ± 0.3	–	–
Mobility						
Diffusion coefficient, 1–10 ns fit, 0–50 ns (nm^2 ns^{-1})	0.365 ± 0.013	3.5	–	0.387 ± 0.044	11	–
Diffusion coefficient, 1–10 ns fit, 50–100 ns (nm^2 ns^{-1})	0.095 ± 0.024	25	–	0.106 ± 0.038	35	–
Diffusion coefficient, 1–10 ns fit, 100–150 ns (nm^2 ns^{-1})	0.073 ± 0.018	25	–	0.066 ± 0.022	33	–
Hexanes moving more than half the box, 150 ns	92 ± 5	–	–	94 ± 4	–	–
Median net displacement, 150 ns (nm)	10.6 ± 1.8	–	–	11.1 ± 2.0	–	–
Distinct partners within 1.5 nm (of 99)	99.0 ± 0.0	–	–	99.0 ± 0.1	–	–

Runs 1–3 are the three original replicas; the ten runs add replicas 4–10, built and run with the same settings (n = 10 unless stated). Per-run values for all ten runs are given in Supplementary Data 1. CV is the coefficient of variation across runs (s.d./mean). R is the between-run standard deviation divided by the mean single-run standard error from five 1-ns blocks of the window (Methods); for the potential-energy rise the single-run error combines the 0–5 and 145–150 ns windows. R is given for window quantities only and is undefined (–) where a quantity did not vary at all. Milestones are the first times after which the 0.5-ns running mean stays above the threshold, and the peak number of clusters is taken from the same running mean. Surface quantities are computed with periodic images except those marked [a], which are computed without them, for comparison. [b]Defined by the last droplet fusion of each run (a merger of two clusters of at least ten molecules, timed from when it held for 0.2 ns), so n = 9 for the ten runs: in run 6 a single droplet grew to its final size without a droplet fusion. The decomposition of the potential-energy rise uses saved full-system frames and is given for runs 1–3 only.

Table 3 | Force-field and time-step checks.

Quantity	Conditions	This work	Lipid14 (ref. 47)	Experiment
Liquid hexane, 150 molecules				
Density (g mL^{-1})	298.15 K, 4 fs, HMR	0.6361 ± 0.0003	0.6363	0.655 (ref. 27)
Density (g mL^{-1})	298.15 K, 2 fs	0.6354 ± 0.0001	–	0.655 (ref. 27)
Density (g mL^{-1})	310 K, 4 fs, HMR	0.6210 ± 0.0002	–	–
Density (g mL^{-1})	310 K, 2 fs	0.6207 ± 0.0002	–	–
Enthalpy of vaporisation (kJ mol^{-1})	298.15 K, 4 fs, HMR	28.28 ± 0.08	28.54 ± 0.1	31.56[a]
Enthalpy of vaporisation (kJ mol^{-1})	298.15 K, 2 fs	27.95 ± 0.08	–	31.56[a]
Enthalpy of vaporisation (kJ mol^{-1})	310 K, 4 fs, HMR	27.40 ± 0.11	–	–
Enthalpy of vaporisation (kJ mol^{-1})	310 K, 2 fs	27.36 ± 0.12	–	–
One hexane in 1,065 waters, 4 fs, HMR				
Hydration enthalpy, gas → water (kJ mol^{-1})	298.15 K	−35.0 ± 1.3	–	≈ −31.5[b]
Hydration enthalpy, gas → water (kJ mol^{-1})	310 K	−31.4 ± 1.4	–	–
Transfer enthalpy, liquid → water (kJ mol^{-1})	298.15 K	−6.7 ± 1.3	–	≈ 0[c]
Transfer enthalpy, liquid → water (kJ mol^{-1})	310 K	−4.0 ± 1.4	–	positive[c]
Partial molar volume (cm^3 mol^{-1})	298.15 K / 310 K	117.1 ± 0.9 / 116.8 ± 1.1	–	–

Means ± standard errors over 20 ns (liquid, 10 blocks) or 199 ns per box (hydration, 20 blocks). Transfer enthalpy = hydration enthalpy + enthalpy of vaporisation at 4 fs. Lipid14 values are from ref. 47 (298.15 K). [a] Experimental value as compiled in ref. 47. [b] Minus the experimental enthalpy of vaporisation, taking the transfer enthalpy as zero. [c] Liquid hydrocarbons transfer into water with zero enthalpy near 22 °C and a positive heat-capacity change, so with a positive enthalpy above that temperature[34].

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Data availability

Hexane coordinates (every 10 ps), full-system coordinates (every 500 ps), energy logs, analysis outputs and the source data for every figure and table will be deposited at [repository, DOI to be added].

Code availability

All scripts will be deposited at [repository, DOI to be added]. These comprise the system builder (build_hexane.py, with the OpenMM template hexane.xml), the simulation driver (run_md.py) and the SLURM job scripts, including 06_extend.sbatch, 07_water_control.sbatch and 08–13 for the controls and replicas. They also include the analysis scripts (analyze.py, mobility.py, sasa_pbc.py, interactions.py, pmf.py and snapshots.py), the force-field checks (liquid_hexane.py and hydration_enthalpy.py) and the scripts for this paper (hex_droplet_pbc.py, random_baseline_trials.py, hex_sasa_pbc.py, hex_energy_split.py, hex_energy_summary.py, hex_exchange.py, hex_fusions_all.py, hex_lastpair.py, hex_waiting.py, hex_gauche.py, hex_mobility_windows.py, hex150_stats.py, hex_dt2.py, hex_ten.py, hex_render.py, check_join.py, paper_figs_v3.py, paper_figs_v4.py, paper_figs_v5.py, stats10.py, run_all10.sh, make_table2_v3.py and control_check.py).

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Author contributions

[To confirm.] M.A.I. designed and ran the simulations, performed the analysis and wrote the manuscript. E.J. supervised the project and edited the manuscript.

Competing interests

The authors declare no competing interests. [To confirm.]